# Behavioral Responses to Taxation of Inherited Property

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**Abstract**

Whether inheritance taxation reduces wealth inequality and raises tax revenue hinges on households' behavioral responses. We examine responses to California's Proposition 19, which eliminated property tax benefits for most inherited homes. Using data from San Francisco and Los Angeles counties, we find that households accelerated inter-vivos property transfers to their children by 13 to 18 months, timing transfers to occur between the policy's announcement and implementation. This response is concentrated among wealthy, white households with higher-value homes, undermining the policy's redistributive and revenue-generating objectives. We also find that properties transferred immediately before implementation are less likely to be sold within four years, consistent with our theoretical framework predicting that accelerating transfers is most beneficial when heirs hold properties longer. Finally, we estimate that accelerated transfers resulted in tax revenue losses of $1.7 to $2.1 billion in Los Angeles and San Francisco alone.

# 1 Introduction

Inheritance taxation is typically utilized to reduce the persistence of wealth inequality across generations and increase government revenues. However, the effectiveness of such a tax critically depends on behavioral responses to the tax. If individuals can shift the timing of inheritance transfers, differential tax treatment for transfers in life versus at death can induce substantial tax avoidance behavior. This tax avoidant behavior can undermine both the redistributive and revenue-increasing goals of the tax—in addition to distorting optimal household choices.

This paper documents the behavioral response to an impending tax increase on inherited property transfers, leveraging a natural experiment created by California's Proposition 19. Under California property tax law, a property's taxable or assessed value is equivalent to the property's purchase price plus an annual 2% increase unless it is sold, at which point it is reassessed to its current market value. Thus, for long-held properties, California's property tax system creates a large wedge between a property's taxable value and its market value, since a property's market value typically increases at a much faster rate than 2% per year. As a result, long-time homeowners enjoy very low, below-market property tax burdens. Prior to the implementation of Proposition 19, a child-heir inherited a property's below-market taxable value along with the property. Proposition 19 eliminated this benefit, mandating reassessment to a property's current market value upon transfer, thus significantly increasing property tax burdens for heirs. In sum, post-Proposition 19 inheritance transfers are treated as sales rather than pre-Prop 19 continuous ownership.

We show that households significantly re-timed property inheritance transfers to avoid the impending property tax increase. Due to the delayed implementation of Proposition 19, a transition window was created for homeowners to transfer their property to an heir before the law took effect. We use data on property transfers in two populous California counties, Los Angeles County and San Francisco County, and find a significant increase in property inheritance transfers within the transition window. We estimate that the policy change accelerated 13-18 months of property inheritance transfers, and cost the government $3.8 billion in expected property tax revenue due to re-timed transfers. We also find that the behavioral response was concentrated among wealthy, white households with higher-value homes. The behavioral response thus undermined both of the standard intentions of inheritance taxation, namely increasing revenue and redistribution, at least temporarily.

To understand potential interactions among incentives and disincentives to accelerate property transfers, we propose a theoretical framework to guide the transfer decision of parent-homeowners. Capital gains taxation is a crucial consideration in our theoretical framework, since inter-vivos inheritance transfers are taxed differently than inheritance transfers occurring at death. Capital gains liability is triggered when an asset is sold, and the amount of tax liability is based on the asset's appreciation since its purchase. However, assets inherited at death receive a step-up in basis, meaning that the heir is only liable for capital gains on the asset's appreciation since the decedent's death—the tax basis is stepped up to the asset's value at death. This step-up in basis is not applicable to inter-vivos inheritance transfers, meaning that capital gains taxation disincentivizes accelerated property transfers, while potential property tax savings incentivizes accelerated transfers. Additionally, parents may be disincentivized from an accelerated transfer by a desire to maintain full legal control of their property. We thus incorporate three dimensions of decision-making into our theoretical framework: property tax savings, capital gains tax liability, and parental asset control. Using our model, we show that the transfer decision depends crucially on the expected death date of the parent and the length of time the child expects to hold the inherited property. We find that accelerated transfers most benefit families in which 1) the child plans to hold the property for many years, 2) the parent is closer to death, and 3) the property has not appreciated significantly.

We next test the predictions of our theoretical framework empirically. First, we examine sale probabilities for properties transferred before and during the transition window, and find that properties transferred during the transition window are less likely to be sold within four years of transfer in the San Francisco sample, but not in the Los Angeles sample. Thus, San Francisco heirs are behaving consistently with our model prediction, but Los Angeles heirs are not. Second, we investigate transfer behavior around the capital gains exemption threshold, and find no significant changes in transfer behavior around the threshold. This is seemingly inconsistent with our theoretical prediction that higher capital gains liability would decrease transfer activity. However, if the parent believes that the child will never sell the inherited property, capital gains liability will be zero, allowing property tax savings to dominate. Finally, we use the model to estimate tax revenue losses to the government, which are equivalent to the property tax savings obtained via accelerated inheritance transfers. We estimate a tax revenue loss of $1.68 billion in Los Angeles County, equivalent to approximately 10% of 2021 property tax revenue in Los Angeles County. In San Francisco County, we estimate a loss of $2.13 billion, equivalent to approximately 70% of 2021 property tax revenue in San Francisco County.

## 1.1 Literature

A large literature examines how inherited wealth transfers respond to changes in tax policy (for a review, see Schratzenstaller (2024)). This research primarily focuses on financial assets, leaving transfers of tangible assets relatively understudied. Tangible assets—particularly durable capital goods—may be especially sensitive to taxation due to their nearly infinite intertemporal elasticity of investment (House and Shapiro (2008)). Thus, inheritance taxation of tangible assets might induce larger behavioral responses than inheritance taxation of other assets. Additionally, real estate typically constitutes a significant portion of parental wealth. This makes the transfer decision consequential for the wealth accumulation of the parent, but also implies that any potential inheritance tax determined by a property's value will be quite significant.

This paper contributes to the literature on how households adjust inter-vivos transfers in response to tax rate changes. While studies suggest that households do not fully utilize tax-preferred gifts to minimize taxes even when estate taxes rise (McGarry (2013); Joulfaian and McGarry (2004); Bernheim et al. (2004)), other research shows large gifts exhibit high short-run elasticities when tax changes are substantial and salient (Ohlsson (2011); Escobar et al. (2019); Locks (2024)). Escobar et al. (2019) find a large elasticity of $e = 22$ for a policy that allowed Swedish heirs to contribute some of their inheritance to their children to avoid taxes on inheritance over a certain threshold. Locks (2024) studied a similar setting but examines the transfers of all assets. The author estimates a large short-run elasticity of wealth transfers of $e = 148$. In contrast, our setting studies the transfer of a particular kind of asset: property. We estimate a much larger elasticity ($e =$) for the intergenerational transfers of residential and commercial property. Since property is a long-lived, high-value asset, our estimates are likely to exceed those for more liquid or short-term assets (House and Shapiro (2008)).

In addition, we contribute to the broad literature on behavioral responses to taxation on intergenerational transfers. Previous research have utilized a notched or kinked tax schedules to quantify the responses to a variety of these tax changes, such as those on estate size at death, wealth accumulation, and taxable wealth transfers (for example, Kopczuk (2013); Goupille-Lebret and Infante (2018); Glogowsky (2021)). Our study focuses on the intergenerational transfer of property and leverages a quasi-experimental policy setting and bunching design.

This paper also contributes to the literature on the effects of local property taxes on the housing landscape like average turnover and prices. In particular, we focus on California's unique property exemptions. Some studies have found that Proposition 13's tax-advantages for longterm homeowners decreased housing turnover and increased housing prices (Wasi and White (2005); Ferreira (2010); İmrohoroğlu et al. (2018)). The California Legislative Analyst's Office reports that higher income households tend to own high valued homes, and thus, receive greater tax relief (Taylor (2016)). Baker (2024) finds that Proposition 13's tax advantage to landlords passes through in the form of lower rent for renters. This paper documents the effects of a change to Proposition 13 and subsequent rules on transfer behavior and inter-generational property turnover.

Lastly, this paper contributes to recent evidence on the regressivity of property taxes, and studies of the disproportionate tax burden on racial and ethnic minorities (Avenancio-León and Howard (2022); Amornsiripanitch (2020)). This paper documents the heterogeneous effects of a property tax change on different racial and socioeconomic groups.

The remainder of the paper will proceed as follows. Section 2 provides background information on the policy change and describes the data. Section 3 describes a theoretical framework that predicts parental transfer choices. Section 4 presents our empirical results and tests the predictions of our model. Section 5 discusses the implications of accelerated property transfers for inequality and tax revenue. Section 6 concludes.

# 2 Background & Data

## 2.1 Background

Enacted in 1978, California's Proposition 13 caps the property tax rate at 1% of assessed value and limits annual increases in assessed value to 2%. Reassessment occurs only when a property is sold, materially improved, or experiences a change in ownership.[1] As a result, property taxes are based on purchase price, or base-year value, rather than current market value. Because housing prices have historically appreciated faster than 2% per year, tax liabilities rise more slowly than market values, lowering effective property taxes for longer-tenured homeowners. For example, a home purchased in 2000 for \$500,000 and appreciating to roughly \$1.2 million by 2020 would have a taxable value of about \$743,000 under Proposition 13, or 62% of market value.

Since Proposition 13, California has adopted additional measures extending limits on property tax reassessment. Two amendments exempt certain intergenerational transfers from reassessment.

[1] Properties held in trust are also subject to reassessment when ownership changes within the trust.

Under these provisions, children inheriting a parent's or grandparent's principal residence, either inter vivos or at death, could retain the property's base-year assessed value and thus its lower effective tax rate.[2] More broadly, parents could transfer up to $1 million in assessed value of non-primary property without reassessment, a limit that was rarely binding.[3]

These provisions conferred substantial tax benefits on the heirs of California homeowners and widened property tax differentials across otherwise similar properties. In the example above, a child inheriting a home worth $1.2 million would pay $7,430 in annual property taxes, rather than $12,000 absent the exemption.

However, in November 2020, California voters approved Proposition 19, which largely eliminated intergenerational reassessment exemptions to fund expanded property tax relief for wildfire victims and older homeowners. Under Proposition 19, inherited primary and secondary residences are generally reassessed at market value, as in an arm's-length sale, with one exception: a $1 million *market-value* exclusion for transfers of a parent's primary residence to a child who also uses the property as a primary residence, which very few households claim after the policy change. Although Proposition 19 passed on November 3, 2020, it did not take effect until February 16, 2021, creating a brief window in which families could transfer property under the prior regime.

## 2.2 Data

We compile data from public sources, including familial transfer claims from county assessor's offices, property tax data on taxable values and home characteristics, and Zillow market values. We link these data to create a panel of San Francisco and Los Angeles properties that captures transfers, tax assessments, and market values.

**Transfer and Tax Data.** We use public-records requests to assemble data on familial transfer claims from two county assessors' offices: San Francisco County (2013-2021) and Los Angeles County (1987-2021). These records report each property's address, assessor parcel number, and transfer date (or claim filing date). The San Francisco data also indicate whether the

[2] To file a claim, the parent submits a short form listing basic identifying information and the signatures of both parent and child. Supporting documentation, such as a transfer document, will, or trust, may be required. Claims are filed with the county assessor and may be submitted in person or by mail at no cost.

[3] Because the $1 million exemption applied to *assessed* value, it bound only when purchase price plus 2% annual assessment growth exceeded $1 million. Average assessed value in 2020 was $929,227 in San Francisco County and $567,626 in Los Angeles County, both below $1 million. Averages for ever-transferred homes are even lower, $735,005 and $308,129 respectively, suggesting very few homes ripe for transfer would be affected by the $1 million exemption threshold.

property was the parent's primary residence. In Los Angeles, we infer primary-residence status from the presence of a nonzero homeowners' exemption.[4] We merge these claims with restricted assessor roll data, which are compiled annually and report assessed value and basic housing characteristics, including bedrooms, bathrooms, and square footage. The assessor data also include owner names, which we use to impute homeowner race.[5]

**Market Value Data.** Because properties are not reassessed annually by the state, we use publicly available Zillow data to measure market values. For San Francisco County, these data cover the universe of properties from 2012 to 2022. For Los Angeles County, coverage spans 2014 to 2022 for properties that were transferred at least once between 1987 and 2021.

Because Zillow coverage is incomplete for transferred properties in San Francisco, we impute missing market values in two steps. First, using properties with observed Zillow values, we estimate a panel model with property fixed effects and tract-specific linear time trends. For properties observed in some years but not others, we use the estimated property effect and tract trend to predict missing annual market values.

Second, for properties with no observed Zillow values during the sample period, we impute the missing property effect and tract trend using nearest-neighbor matching. For each unmatched property, we identify the five geographically nearest properties in the estimation sample and construct similarity weights based on standardized differences in housing and neighborhood characteristics, including distance, tract median income, bathrooms, bedrooms, rooms, units, square footage, lot area, and number of stories. We then use these weights to form weighted averages of the matched properties' estimated fixed effects and tract trends, which generate out-of-sample predictions for market values. If any matched neighbor is missing one or more characteristics, we instead assign equal weights across the five neighbors. We winsorize the final predicted market values at the 5th and 95th percentiles. More details on the construction of this variable can be found in Appendix C.1.

We also use CoreLogic's zipcode-level House Price Index to impute current market values for homes in our sample not covered by publicly available Zillow data. These indices exist from 1975–present.

[4] In California, the homeowners' exemption applies to owner-occupied primary residences and reduces assessed value by $7,000 per the California State Board of Equalization.

[5] We impute race using the open-source `rethnicity` package (Xie and Fangzhou (2022)). Because some properties, especially transferred properties, are held in trusts or limited liability companies, owner names do not always identify households cleanly. To reduce this problem, we remove terms such as trust," LLC," and "family" from ownership records before analysis.

**Sale Data.** We use property sale dates from CoreLogic's Real Estate Database to determine the timing of sales after a parent-child property transfer. The CoreLogic data contain a property's address, parcel number, property characteristics, and sale records. We merge these records with the county assessor's parent-child transfer claims data using a property's parcel number and/or street address.

**Sample Restrictions.** Our analysis includes only properties classified as "residential," explicitly excluding properties that are used for commercial purposes. Additionally, we limit the San Francisco sample to properties that have available data on market value and geolocation due to the nearest neighbor matching procedure. However, this only removes 0.6% of unique properties in the San Francisco dataset.

# 3 Model

We model the parent's decision of when to transfer a property to their child. Relative to a transfer at death ($t = d$), an accelerated transfer before Proposition 19 takes effect ($t = 0$) has three consequences. First, it allows the child to retain the property's pre-existing assessed value, thereby avoiding reassessment and lowering future property tax payments. Second, it causes the child to forgo the step-up in basis at death, increasing capital gains liability if the child later sells the property. Third, it reduces the parent's control over the property prior to death. We capture this last margin with a reduced-form control cost rather than modeling intra-household bargaining explicitly. We assume that the child sells the property in period $t = s > d$ regardless of the transfer date. For simplicity, we also assume a perfectly dynastic setting in which the parent values their own and their child's utility equally.

We do not consider transfers after Proposition 19 but before death in our framework. Because such transfers no longer avoid reassessment, they deliver no property tax benefit relative to a transfer at death, while still requiring the parent to relinquish control before death and potentially causing the child to forgo the step-up in basis. They are therefore strictly dominated by a transfer at death.

If the property is transferred before Proposition 19 takes effect, the child retains the parent's tax basis and continues to pay property taxes on the capped assessed value. If instead the transfer occurs at the parent's death, at time $d$, the property is reassessed to its market value at death, denoted $MV_d$, and property taxes rise accordingly. Let $TV_0$ denote the property's taxable value at the time of the potential pre-Proposition 19 transfer, $t = 0$; let $s > d$ denote the date at which the child sells the property; let $\tau_p$ denote the property tax rate; and let $\rho$ denote the discount factor. Then the present value of the property tax savings from accelerating the transfer is

$$\Delta PT(d,s) = \sum_{t=d}^{s} \rho^t \tau_p \left[ MV_d \cdot 1.02^{t-d} - TV_0 \cdot 1.02^t \right]. \tag{1}$$

Accelerating the transfer also increases capital gains taxation if the child later sells the property, because the child forgoes the step-up in basis at death. Let $\tau_{cg}$ denote the capital gains tax rate and $MV_p$ the parent's original purchase price. The additional capital gains tax liability from early transfer is then

$$\Delta CG(d,s) = \rho^s \tau_{cg} (MV_d - MV_p). \tag{2}$$

If the child never sells the property, this term is zero.

Finally, let $\kappa_d$ denote the non-tax cost of transferring the property before death, which may vary with the parent's remaining lifespan. The parent prefers to accelerate the transfer when the property tax savings exceed the sum of this control cost and the additional capital gains tax liability:

$$\Delta PT(d,s) > \kappa_d + \Delta CG(d,s). \tag{3}$$

Equation (3) captures the central trade-off in a tractable way. Since the left-hand side of Equation (3) is increasing in $s$ (and decreasing in $d$) and the right-hand side is decreasing in $s$ (and increasing in $d$), accelerated transfer is more attractive when reassessment would generate a large increase in property taxes, when the parent is closer to death, and when the child expects to hold the property for a longer period after inheritance. By contrast, it is less attractive when the parent places a high value on retaining control of the property or when forgoing the step-up in basis generates a large capital gains tax burden. See Appendix C.2 for a detailed breakdown and a numeric example.

We do not incorporate federal gift and estate taxation into the baseline model. In our setting, these taxes are unlikely to affect most transfers because only a small share of properties exceed the relevant exemption threshold, and we lack information on households' broader wealth portfolios and prior lifetime gifts.[6] We therefore treat gift and estate taxation as a secondary

[6] Transfers of property trigger a federal gift tax liability on the giver (in this case, the parent); however, this

margin outside the baseline framework.

# 4 Empirical Results

As previously discussed, California homeowners accrue substantial property tax benefits by remaining in the same home over time. The longer homeowners remain, the greater the gap between the market and taxable values of their properties, resulting in progressively lower effective property tax rates.[7] Thus, transferring a property's taxable value to a child—as was allowed before Proposition 19—extends these tax advantages to the next generation of homeowners.

Table 2 provides summary statistics on market values, taxable values, and home characteristics for all homes in San Francisco County, as well as a sample of homes that were ever transferred in our sample in both San Francisco and Los Angeles Counties. Within this sample, average market value exceeds average taxable values by more than $1 million in San Francisco and $500,000 in Los Angeles, resulting in very low effective property tax rates of approximately 0.4–0.6%. The average home is generally held for more than 20 years; however, those who have ever transferred their homes have typically held onto their property two to four years longer. Additionally, homes in San Francisco tend to be higher-value, larger, and located in Census tracts with higher average household income compared to homes in Los Angeles.

These substantial differences between market and taxable values create significant stakes for inheritors in terms of potential property tax savings. Figure 1 reports average potential annual tax savings for properties transferred before the Proposition 19 deadline within the sample of ever-transferred homes. Potential tax savings are equal to the difference between the tax liability of a property's market value and the tax liability of a property's taxable value in 2020. Panel (a) shows that inheritors of median-value homes in San Francisco County would save $11,779 annually by avoiding reassessment through pre-deadline transfers. Conversely, transferring after the deadline implies an annual tax increase of at least $11,779

tax only applies once an individual's cumulative lifetime gift exceeds the exemption threshold of $12 million per beneficiary. Consequently, this tax primarily affects wealthy households or those holding highly valued assets. Approximately 0.5% properties in San Francisco and Los Angeles counties in 2020 have market values that surpass this threshold, indicating that, based solely on property transfers, our sample remains largely unaffected by federal gift taxes. Nonetheless, households with substantial wealth may strategically time property transfers in consideration of other lifetime gifts. Because we lack sufficient information to precisely evaluate this behavior, and since the median home value in our sample is approximately $1.6 million in 2020, we omit this potential channel from our analysis.

[7] Effective tax rate = Property tax payments / Property market value.

upon inheritance. In Los Angeles County, where homes generally have lower values and slower market appreciation, the potential annual tax savings remain significant, averaging \$4,869. These figures underscore the substantial monetary implications of transferring a property before the Proposition 19 deadline.

## 4.1 Transfer Claims

Using monthly parent-to-child property transfer claims from San Francisco and Los Angeles counties, we document a large increase in the number of transfers that occurred after the policy announcement during the four-month transition window.

Figure 2 reports the number of monthly parent-to-child property transfers in San Francisco and Los Angeles counties. Property transfer claims rose dramatically during the transition window in both counties. In Figure 2 Panel (a), we find that an additional 18 months of transfer claims occurred in San Francisco during the transition window. In Los Angeles, Figure 2 Panel (b) shows that an additional 13 months of transfer claims were induced by the policy change. This is equivalent to transferring an additional 1.4% and 0.13% of the total residential housing stock in San Francisco and Los Angeles respectively, suggesting a larger proportional response in San Francisco County.[8]

Additionally, we can separate claims by parental residence status—i.e., if the property is the parent's primary home where they reside, or if the property is the parent's secondary home. Prior to the passage of Proposition 19, the shares of each type of home were relatively equal, with 55%:45% primary to non-primary in San Francisco, and 52%:48% in Los Angeles. Following the policy announcement, the share changes to 39%:61% in San Francisco and 45%:55% in Los Angeles.[9]

To assess policy impacts on tax revenues, Figure 3 documents the market values of transferred properties over time. Panel (a) shows that an additional \$5.5 billion in market value was transferred during the transition window in San Francisco County, while Panel (b) shows that an additional \$4.7 billion was transferred in Los Angeles County. The value of these transfers represents foregone tax revenue that would have been collected under the post-Proposition 19 tax regime.

[8] San Francisco has 193,116 residential housing units, Los Angeles has 2,152,235. See Table 2.

[9] Recall that prior to the implementation of Proposition 19, primary home transfers faced no exemption limit, while secondary homes faced a \$1 million assessed value exemption limit that likely did not bind for most transfers. After Proposition 19, secondary homes and primary-to-secondary transfers receive no tax advantage, while there is a narrow carve out for primary-to-primary transfers.

## 4.2 Mechanisms

In Section 3, we showed that a parent should transfer their property to their child before Proposition 19 took effect if 1) the child did not plan to sell the property quickly; 2) the parent was relatively close to death; 3) the home's appreciation was well below the capital gains exclusion threshold. In this section, we test these model predictions.

We first test whether children sell transferred properties quickly. Figure 4 shows the percentage of properties sold within four years by transfer year. We separate homes transferred during the Proposition 19 transition window from other homes transferred in 2020. The figure provides suggestive evidence that properties transferred during the transition window in San Francisco are less likely to be sold quickly after, but there is no obvious difference in sale horizons for transition window homes in Los Angeles. However, baseline sale probability is much lower in Los Angeles County than in San Francisco County, with only 5% of transferred Los Angeles homes sold within four years on average, compared to over 30% of San Francisco homes.

We then estimate the probability of sale for transferred properties over different horizons, $s$, after the year of transfer, $t$, using Equation 4. The regression includes property-level characteristics, such as number of bedrooms, size, and year of construction, as well as a census tract fixed effect $\mu_c$.

$$Pr(Sale)_{i,t+s} = \mu_t^s + \gamma X_i + \mu_c + \varepsilon_{it} \tag{4}$$

The year of transfer fixed effect, $\mu_t^s$, identifies the differences in probability of sale by year $t+s$ for homes transferred in year $t$, after controlling for location and property characteristics. Figure 5 plots these transfer year fixed effects for sale likelihood by over different horizons after the transfer year. Both figures 4 and 5 suggest that transition window inheritors in the SF sample are less likely to sell the inherited property in the first four years after the transfer, but not in the LA sample.

We cannot empirically test whether parents closer to death were more likely to transfer their properties during the Proposition 19 transition window, because we do not observe parental ages. However, we do see some suggestive evidence that transition window transferors are older. In Table 3, we show that properties transferred during the transition window had been held longer than previously transferred homes. Longer tenancy is likely correlated with parental age, suggesting that these parents may have been older and thus closer to death.

We next test whether properties with appreciation values below the capital gains exemption threshold are more likely to be transferred during the transition window. Figure 6 plots the distribution of properties transferred by each property's estimated nominal capital gains. Properties with capital gains well below the exclusion threshold are unlikely to trigger any capital gains taxation upon sale and thus would be prime candidates to be transferred during the transition window. However, we do not observe meaningful changes in behavior around the capital gains threshold. This suggests that the capital gains implications of an inter-vivos transfer were possibly misunderstood or deemed unimportant, like in the case of the child planning to keep the inherited property indefinitely.

Finally, though this was not included in our model, we show in Section 5.1 that wealthy, white households were more likely to transfer during the transition window. Wealthy households are likely better able to hold properties for longer so as to amass property tax savings and avoid capital gains taxation, which could operate through the time-to-sale channel.

# 5 Implications

Recall that inheritance taxation policies typically have two main goals: reducing inequality and increasing tax revenue. In this section, we show that transition window transfers were mainly undertaken by the wealthy and cost the government significant tax revenue, temporarily undermining both aims.

## 5.1 Heterogeneity

We examine heterogeneity in responses during the transition window (November 3, 2020 – February 16, 2021) and compare these to a control window, defined as transfers occurring one year prior (November 3, 2019 – February 16, 2020). We investigate differences based on property characteristics, income levels, and race.

### 5.1.1 Property Characteristics

Figure 8 plots the density of property transfer claims by the property's effective tax rate. Effective tax rates are equivalent to taxes paid divided by the market value of the property, or the true proportion of the property that is being taxed. The blue density distribution represents the density of claims that occurred during the transition window, while the red density represents the density of claims that occurred during the control window. In San Francisco, the window and control distributions look relatively similar, with one difference being the slightly lower concentration of claims in the right tail and the slightly higher concentration of claims in the middle of the distribution. This suggests that homes transferred during the transition window have somewhat lower effective tax rates. Similarly, in Los Angeles, homes transferred during the transition window have lower effective tax rates, on average. This pattern suggests that homes transferred during the transition window have been held longer, on average, than those in the control window.[10]

We note other differences in pre-policy vs. post-policy announcement transfers. Homes transferred in response to the policy announcement have higher average market values. To demonstrate this visually, Figure 7 plots the monthly average market value of transferred homes over time. We observe a very large increase in average market value ($730,000) for homes transferred during the transition window in Los Angeles, and a smaller but still sizable increase in value for homes transferred in San Francisco ($400,000). This evidence demonstrates that homes transferred during the transition window are significantly more valuable than the typical transferred home.

Table 3 displays the mean differences of other property characteristics. The table further shows that homes transferred in the transition window also tend to be larger, featuring more bedrooms, bathrooms, and square footage compared to homes transferred in Los Angeles County the previous year. Additionally, properties transferred after the announcement had typically been held by the prior owner (the parent) for 5 to 8 years longer than those transferred in the previous period. Finally, Los Angeles homes transferred in the window are from higher income tracts, which is discussed further in Section 5.1.2.

### 5.1.2 Income

In Figure 9, we show the density of claims by income, using median household income for the Census tract of the transferred home. In San Francisco County, transfer claims are concentrated slightly above San Francisco County's median income of $125,000. There are no obvious changes in this pattern when comparing the transfer and control windows.

In contrast, the modal transfer in the control window for Los Angeles County occurred in zip codes with below-median income. During the transfer window, however, the modal transfer in Los Angeles occurs in an above-median income zip code, and there is a large right tail indicating an increase in the number of transfers in high-income zip codes. This pattern suggests that in normal times, transfers were more evenly distributed among income groups,

[10]Or that they have experienced lower market value growth relative to taxable value, but Figure 7 and Table 3 show that these are high-value homes.

while during the window, homes in wealthy neighborhoods made up the majority of transfers.

#### 5.1.3 Race

Figure 10 shows the density of transfer claims by the proportion that are white for each Census tract. In San Francisco, there is no evident disparity between high- and low-minority Census tracts when comparing the window transfer group to the control group. However, but a striking pattern emerges in Los Angeles. In Los Angeles, the control group contains significantly more transfer claims from Census tracts with higher proportions of minorities, while the window group displays the opposite pattern. Many more transfers in the window group occur in majority-white Census tracts, and there is a reduction in claims from high-minority tracts relative to the control group.

Figure A3 confirms this finding in Los Angeles County using the predicted race of the homeowner. The figure shows the proportion of claims belonging to homeowners of different racial groups for the control period and transition window. For claims in the control period, claims were distributed among racial groups more proportionally to Los Angeles demographics.[11] However, claims during the transition window disproportionately originate from white homeowners, with the share from Hispanic owners significantly lower during this period.

#### 5.1.4 Summary

Taken together, evidence from Section 5.1 shows that homes transferred during the transition window were more valuable, longer-held, and more likely to originate from high-income and low-minority Census tracts relative to their control period counterparts. This suggests that the creation of the transition window disproportionately benefited high-income individuals with high-value homes.

### 5.2 Tax Revenue Loss

Section 3 provides a helpful decision rule for accelerated property transfers, where the parent prefers to accelerate the transfer when the property tax savings exceed the sum of the parental control cost and the additional capital gains tax liability:

$$\Delta PT(d,s) > \kappa_d + \Delta CG(d,s).$$

[11] The racial breakdown of Los Angeles County in 2020 was 49% Hispanic, 25% White Non-Hispanic, 7% Black, and 15% Asian, per the U.S. Census.

Helpfully, property tax savings for the homeowner are equivalent to property tax revenue losses incurred by the county. Thus, using the model from Section 3, we can provide an estimate of the tax revenue lost due to accelerated property transfers. We assume that homes transferred during the transition window satisfy Equation (3) reproduced above, which provides a bounded range for $s$ and $d$.

Figure 11 presents a boxplot of estimated tax revenue losses in San Francisco and Los Angeles counties for homes transferred during the transition window. These estimates are calculated over ranges of $s$ and $d$ for which Equation (3) is satisfied.[12] Los Angeles County had an estimated loss of \$0.5-2.75 billion in tax revenue due to accelerated transfers, with the average estimate equal to \$1.68 billion. This is equivalent to approximately 10% of fiscal year 2021 property tax revenue in Los Angeles County. In San Francisco County, the estimates are larger due to the higher proportional response—we estimate a loss of \$2.13 billion, with the distribution ranging from \$0.75-3.5 billion. This is equivalent to approximately 70% of fiscal year 2021 property tax revenue in San Francisco County.[13]

In sum, estimated losses due to accelerated transfers can be a substantial portion of tax revenue. As inheritance taxes are typically implemented in order to increase tax revenue, our findings caution against the creation of transition windows in future tax policy design.

# 6 Conclusion

Tax policies affecting transfers of real property merit particular attention due to distinctive economic characteristics of real assets. Unlike financial wealth, real property is inherently illiquid, indivisible, and typically represents the largest asset in many household portfolios. These features imply that taxes levied upon real property transfers can induce uniquely pronounced timing responses. Moreover, because real estate often comprises a substantial portion of intergenerational wealth, understanding behavioral reactions to real asset taxation is critical to evaluating policy effectiveness in mitigating wealth inequality and raising tax revenue.

We document substantial behavioral responses to an increase in taxation on inherited property, leveraging the passage and delayed implementation of California's Proposition 19 as a natural experiment. We find that households significantly accelerated property transfers in anticipation

[12]Figure A6 shows the full range of estimated tax revenue losses over all values of $s$ and $d$.

[13]In FY2021, property tax revenue totaled \$17.3 billion in Los Angeles County and \$3 billion in San Francisco County.

of higher future taxes, compressing an additional 13-18 months of transfers into the four-month window between the proposition's approval and its effective date.

We employ a theoretical framework to model the tax tradeoffs of an accelerated property transfer: a reduction in property taxation but a potential future increase in capital gains taxation. Our framework suggests that parents should accelerate a property transfer if 1) the child does not plan to sell the property quickly; 2) the parent is relatively close to death; 3) the home's appreciation is well below the capital gains exclusion threshold. We test our model implications on accelerated property transfers, and show that transfer behavior does not appear to be driven by avoiding capital gains liability, but that homes transferred during the transition window are less likely to be sold quickly by the heir.

Further, we show that the behavioral response to the policy change undermined the policy's redistributive and revenue objectives. Households that accelerated property transfers predominantly resided in higher-income and disproportionately white Census tracts, transferring high-value homes to descendants to circumvent future property reassessments. Using our theoretical framework, we estimate that these accelerated transfers cost the government approximately \$3.8 billion in foregone revenue in Los Angeles and San Francisco counties alone.

In sum, we provide novel evidence on behavioral responses to taxation changes by examining property inheritance—a largely unexplored context despite its significant implications for public finance and wealth inequality. Transfers of inherited real property are particularly compelling because real estate represents a substantial and highly illiquid component of household wealth, creating distinct incentives for tax avoidance through strategic timing. Our results illustrate that transitional windows allowing taxpayers to adjust behavior in anticipation of higher taxes predominantly benefit wealthier individuals, temporarily exacerbating inequality, delaying revenue gains, and reducing overall tax revenues.

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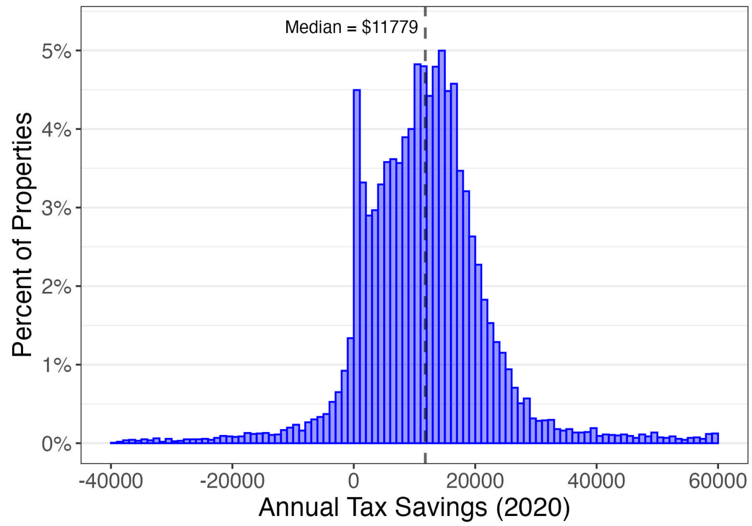
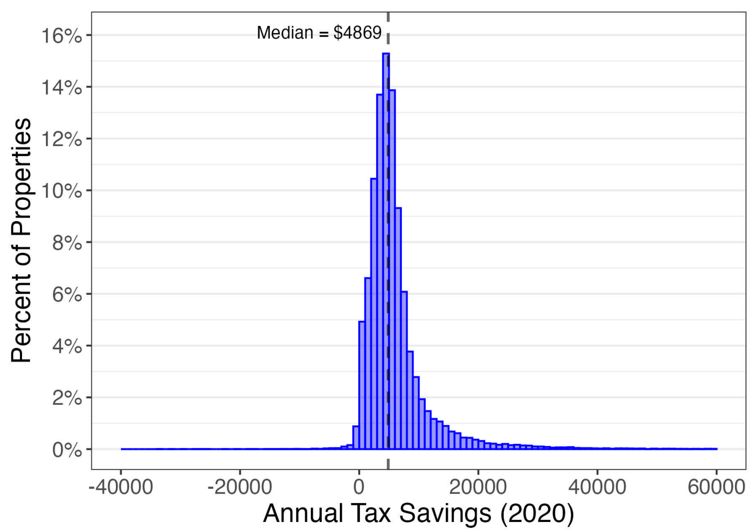
# A Figures

Figure 1: Potential Annual Tax Savings

(a) San Francisco County



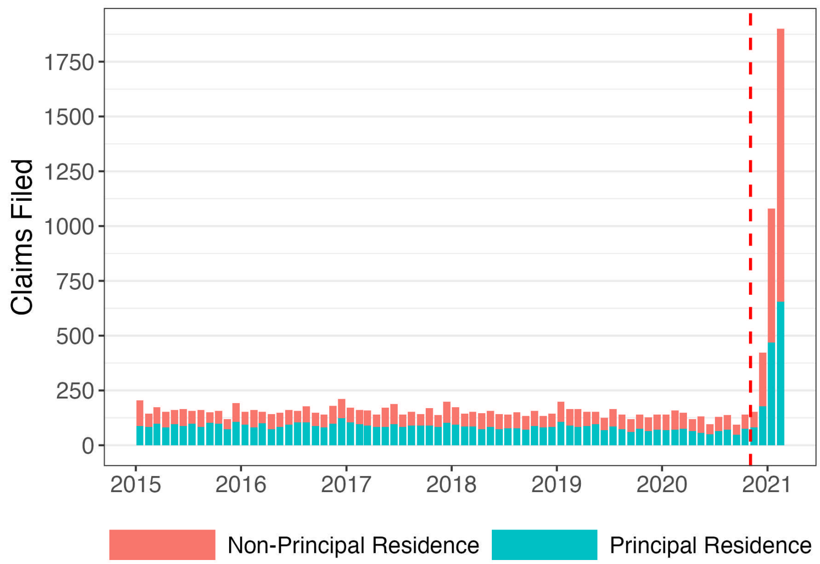

(b) Los Angeles County



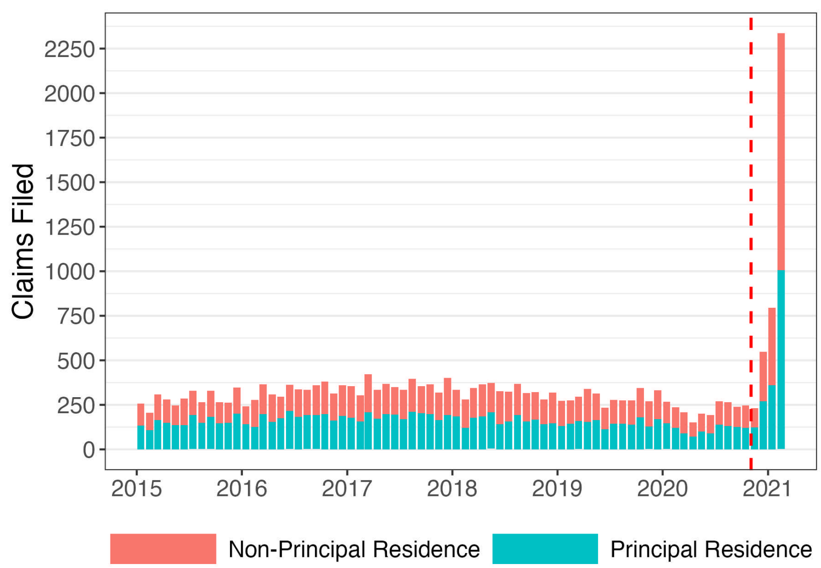

Note: This figure plots a histogram of potential annual tax savings in dollars for the sample of ever-transferred properties in Los Angeles and San Francisco Counties. Potential tax savings are calculated by taking the difference of property tax liability on the market value of the property and property tax liability on the taxable value of the property. Panel (a) plots the values for San Francisco County, and Panel (b) plots the values from Los Angeles County. The vertical dashed line denotes within-county median potential tax savings.

Figure 2: Parent-child transfers in San Francisco and Los Angeles County

(a) San Francisco County



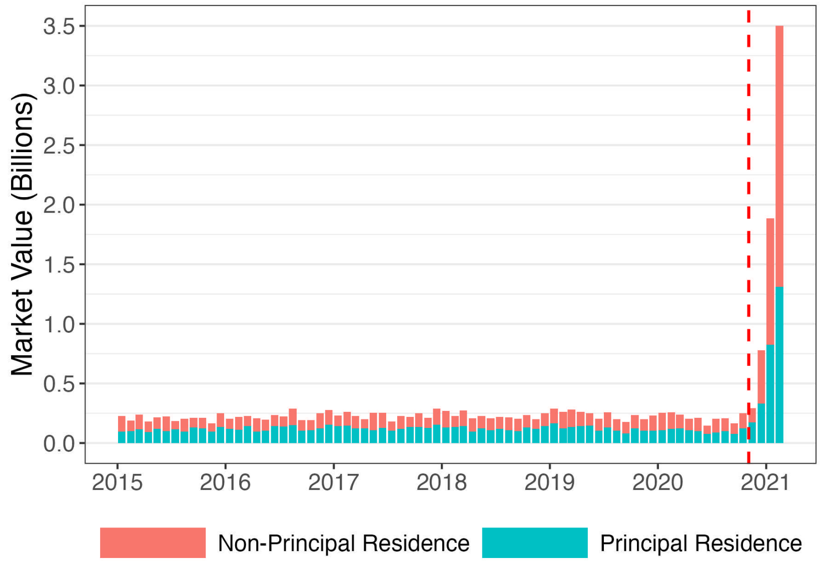

(b) Los Angeles County



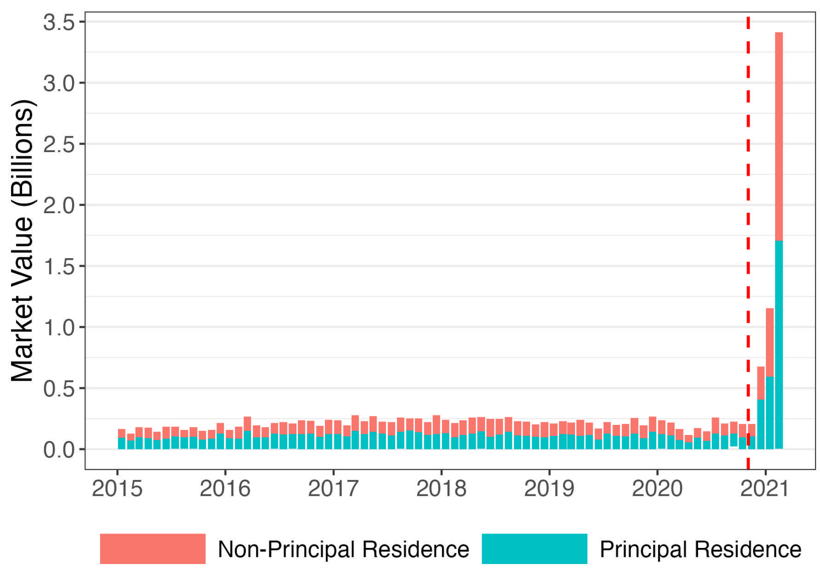

Note: This figure plots the number of unique parent-to-child property transfers by month from 2015 to 2021 in San Francisco County and Los Angeles County. Panel (a) plots the claims from San Francisco County, Panel (b) plots the claims from Los Angeles County. The red bars represent the number of claims for the parents' non-principal or secondary homes and the teal bars represent the number of claims for principal or primary homes. The red dashed line in both panels represents the time of the announcement of the repeal of the parent-to-child reassessment exemption.

Figure 3: Market value of transferred properties in San Francisco and Los Angeles County

(a) San Francisco County



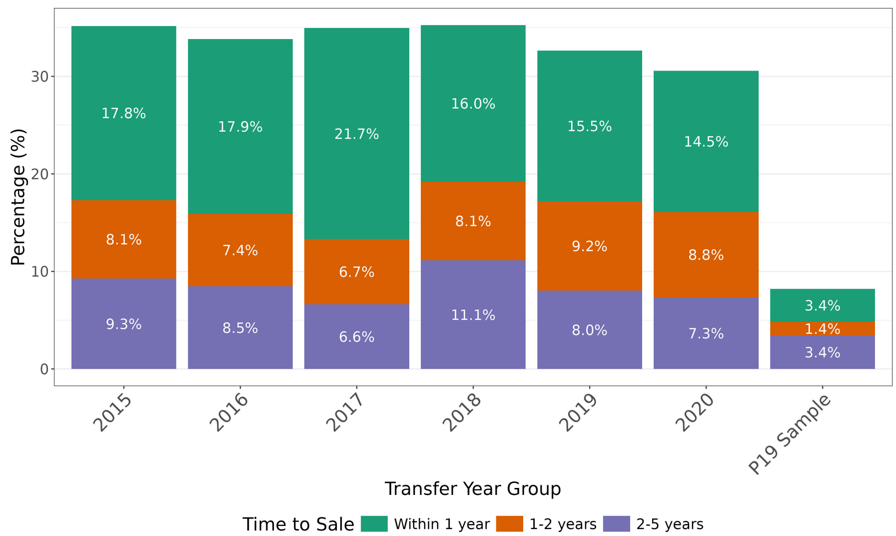

(b) Los Angeles County



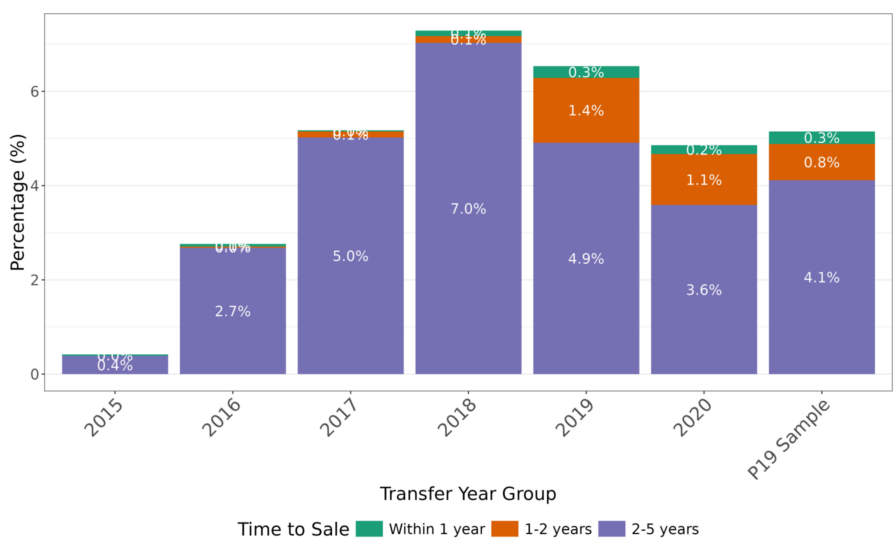

Note: This figure plots the market value of unique parent-to-child property transfers by month from 2015 to 2021 in San Francisco County and Los Angeles County. Panel (a) plots the claims from San Francisco County, Panel (b) plots the claims from Los Angeles County. In both panels, the red bars represent the number of claims for non-principal or secondary homes, the teal bars represent the number of claims for principal or primary homes, and the red dashed line in all panels represents the time of the announcement of the repeal of parent-to-child reassessment exemption.

Figure 4: Percentage of Transferred Properties Sold by Transfer Year

(a) San Francisco County

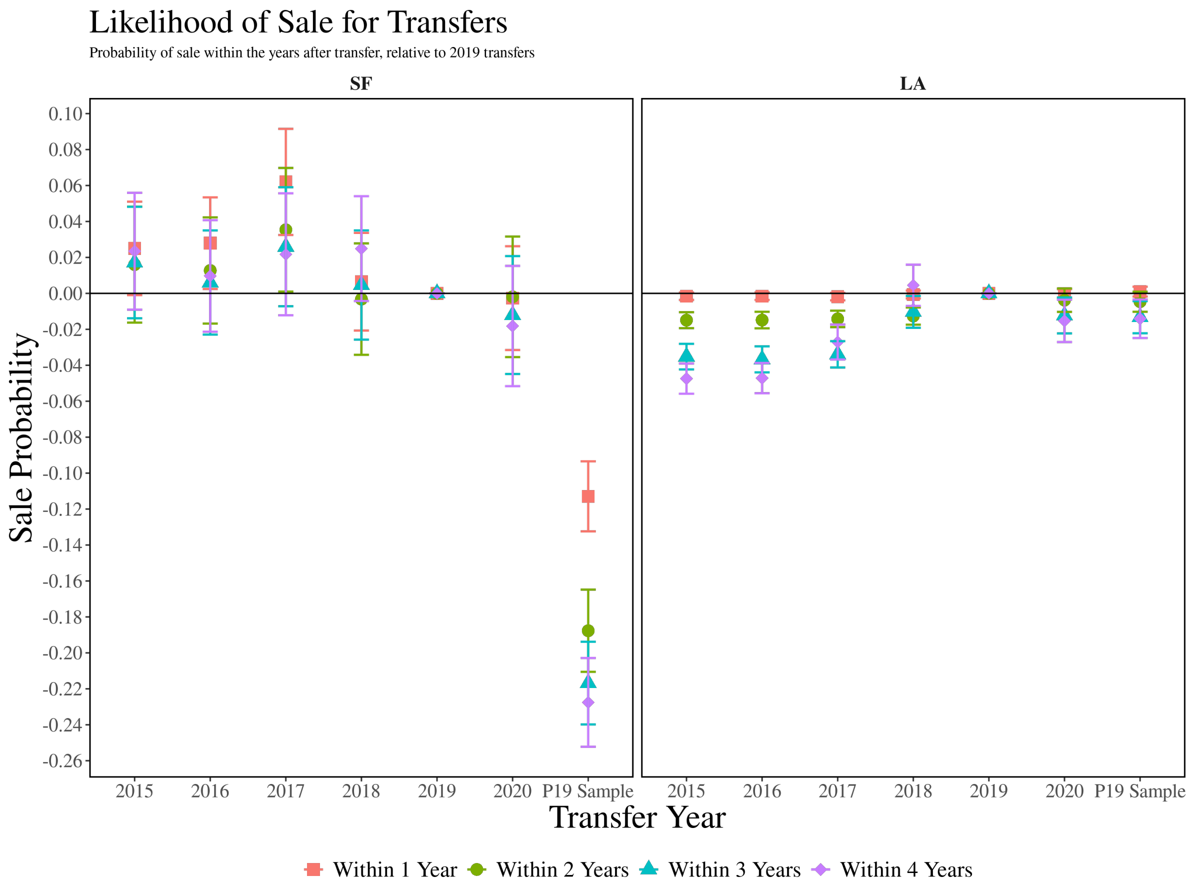



(b) Los Angeles County




Note: This figure plots the percentage of transferred properties by year that are sold within five years for San Francisco and Los Angeles Counties. The green bars denote properties sold within one year, the orange bars denote properties sold between one and two years, and the purple bars denote properties sold after two years but before five years. Panel (a) shows results for San Francisco County, and Panel (b) shows results for Los Angeles County.

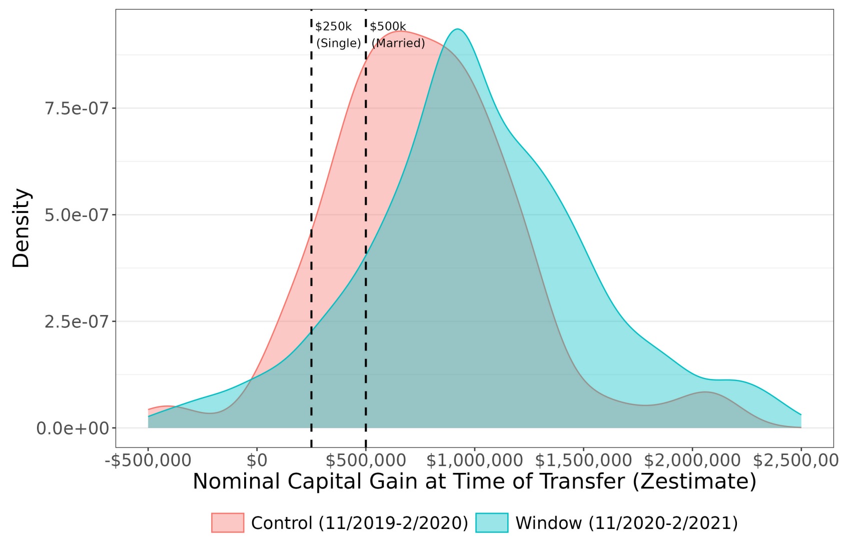
Figure 5: Likelihood of Sale for Transfers



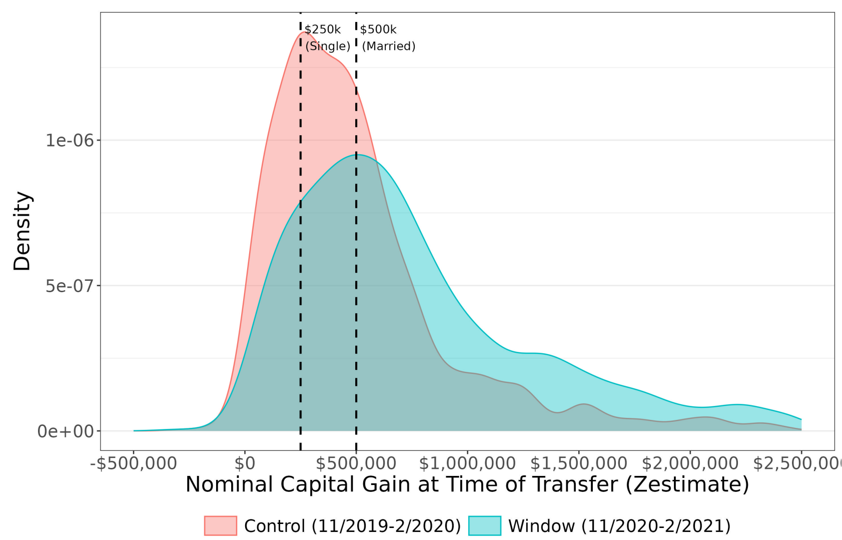

Note: This figure plots estimates of $\hat{\mu}_t^s$ from Equation 4, which measures the difference in probability of sale by year $t+s$ for homes transferred in year $t$. The coefficients are estimated relative to transfer year 2019 and the specification controls for property characteristics (size, number of bedrooms, and year of construction) and tract fixed effects. The left panel plots coefficients for San Francisco County and the right panel plots coefficients for Los Angeles County. Different colored markers represent different horizons to measure the likelihood of sale after the year of transfer. Standard errors are clustered by tract and the error bars show 95% confidence intervals.

Figure 6: Claim Density by Potential Capital Gains

(a) San Francisco County

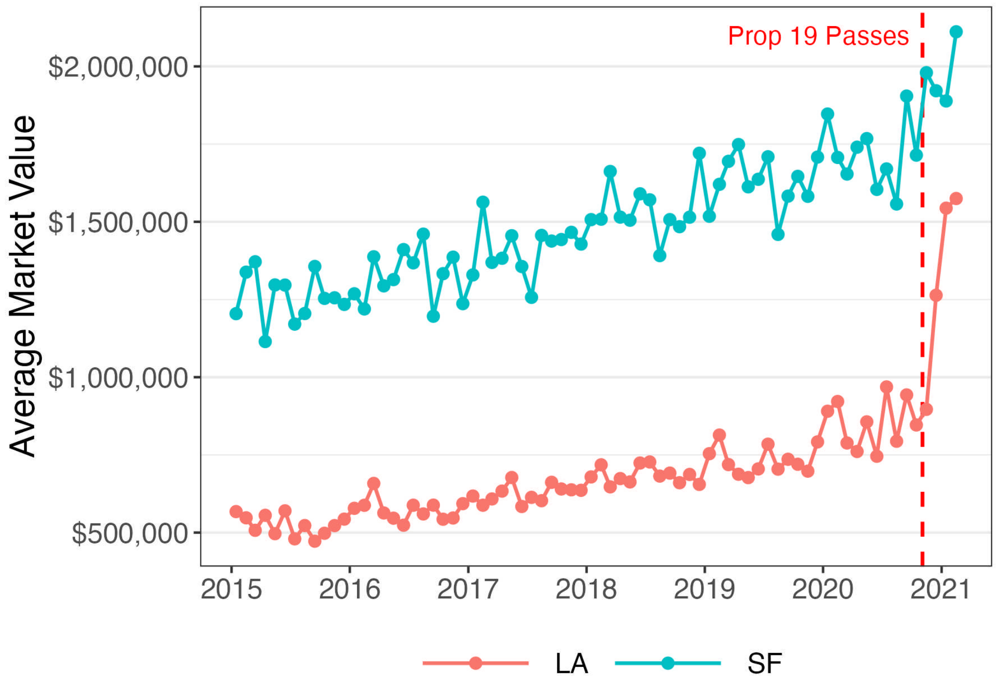



(b) Los Angeles County




Note: This figure plots kernel density estimates of property transfer claims by property-level estimated nominal capital gains. We estimate nominal capital gains by subtracting a property's purchase price from its 2020 Zestimate value. Purchase prices are estimated by deflating a property's taxable value by 2% per year to the property's purchase year. Two groups of transferred properties are compared: properties transferred during the treatment ('Window') period, shaded in blue, and properties transferred during a control period, shaded in red. Panel (a) shows estimates for San Francisco County, and Panel (b) shows estimates for Los Angeles County. The black dashed vertical lines denote the capital gains exclusion threshold for a single homeowner ($250k) and dual homeowners ($500k).

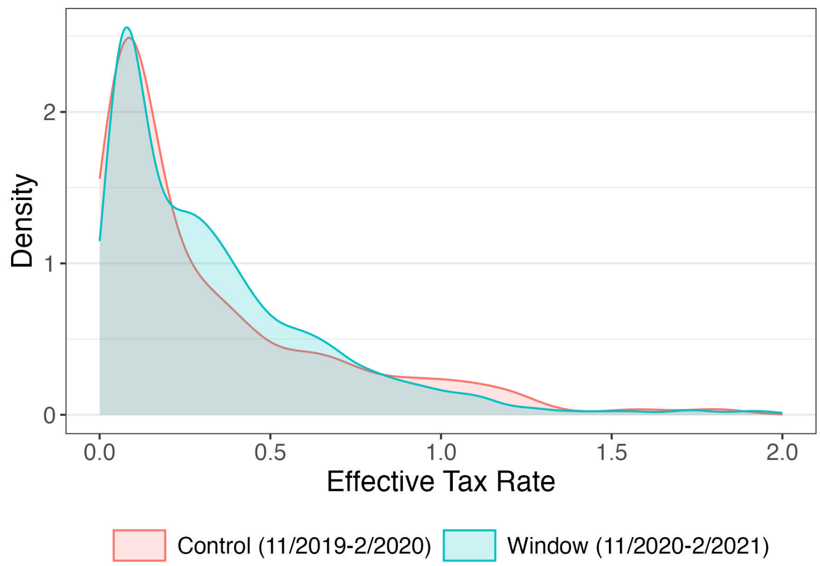
Figure 7: Average Market Value of Transferred Homes Over Time



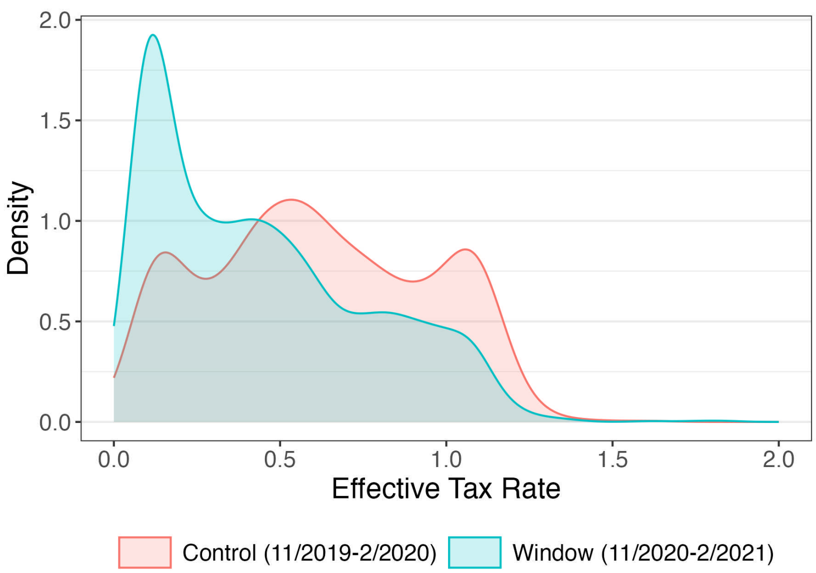

Note: This figure plots the average market value in dollars of transferred homes by month from 2015 to 2021. Los Angeles County market values are represented by the red line and San Francisco County market values are represented by the blue line. The vertical red dashed line represents the time of the announcement of the repeal of parent-to-child reassessment exemption.

Figure 8: Claim Density by Effective Tax Rate

(a) San Francisco County



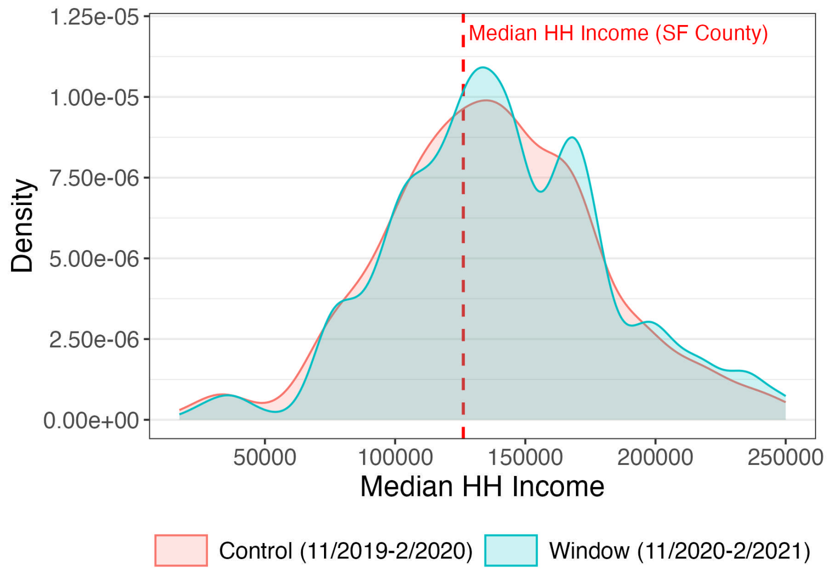

(b) Los Angeles County



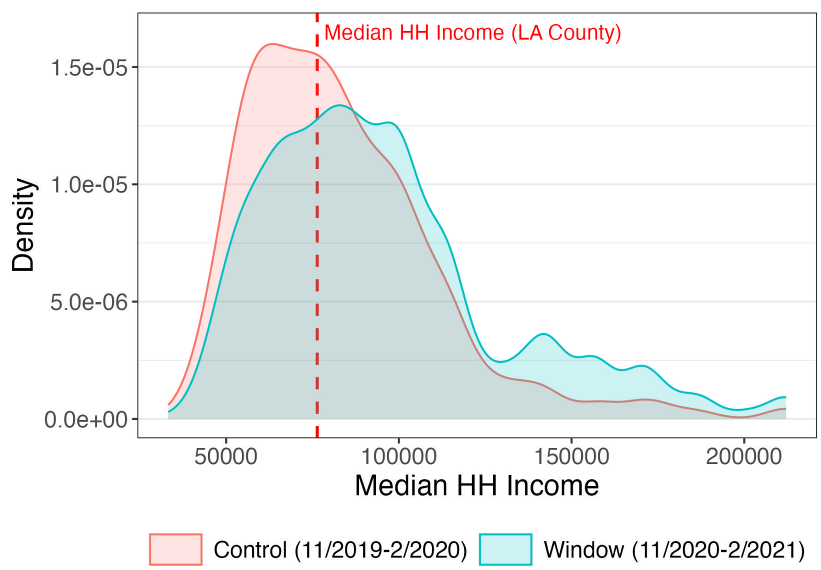

Note: This figure plots kernel density estimates of property transfer claims by the property's effective tax rate. Estimates are divided into two periods: the treatment ('Window') period, shaded in blue, and a control period, shaded in red. Panel (a) shows estimates for San Francisco County, and Panel (b) shows estimates for Los Angeles County.

Figure 9: Claim Density by Census Tract Income

(a) San Francisco County



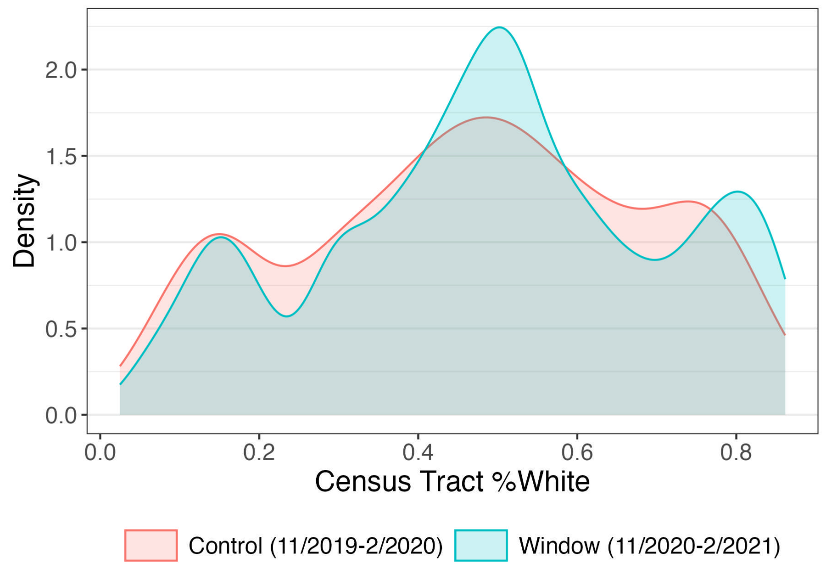

(b) Los Angeles County



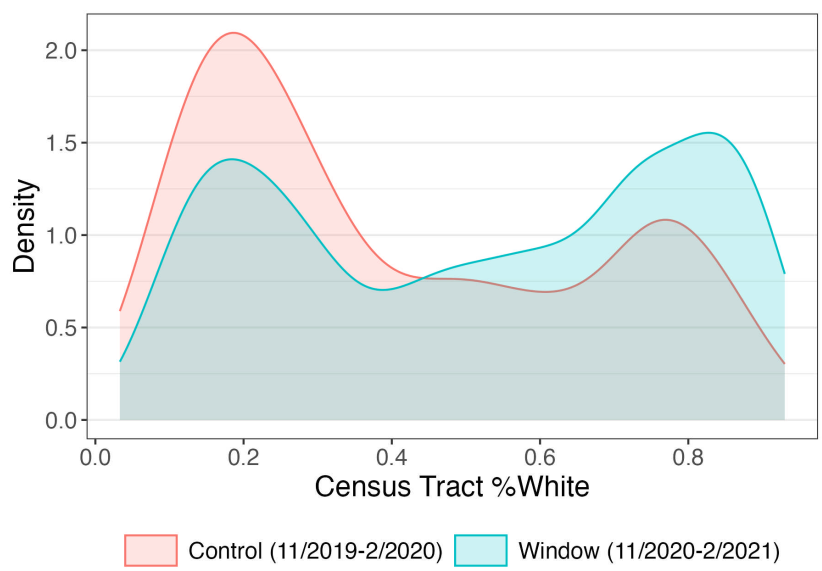

Note: This figure plots kernel density estimates of property transfer claims by the median household income of the property's Census tract. Estimates are divided into two periods: the treatment ('Window') period, shaded in blue, and a control period, shaded in red. Panel (a) shows estimates for San Francisco County, and Panel (b) shows estimates for Los Angeles County. The red dashed vertical lines in both panels denote within-county median household income.

Figure 10: Claim Density by Census Tract Proportion White

(a) San Francisco County

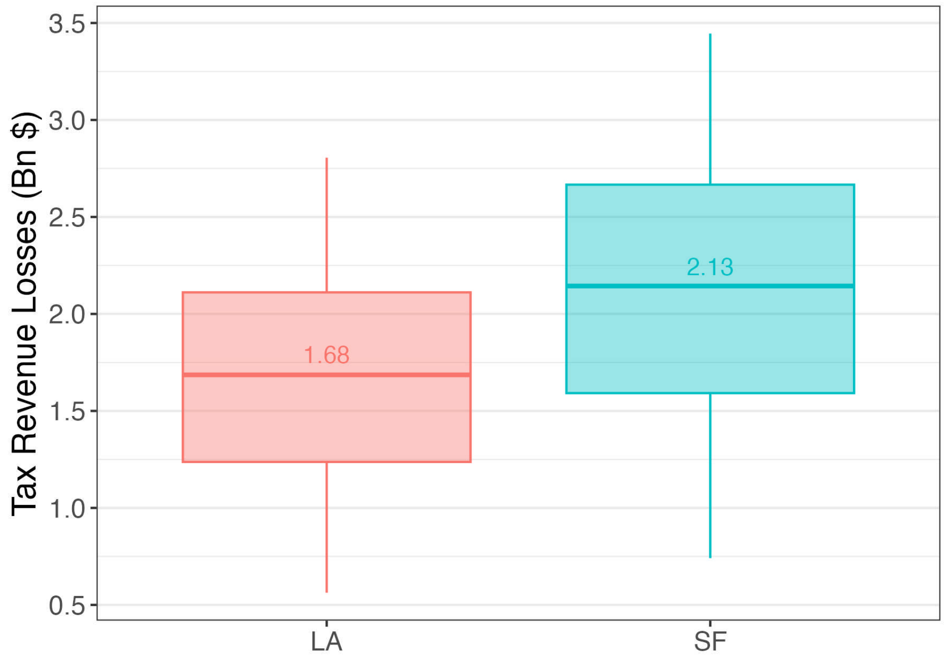



(b) Los Angeles County




Note: This figure plots kernel density estimates of property transfer claims by the percentage of white residents in the property's Census tract. Estimates are divided into two periods: the treatment ('Window') period, shaded in blue, and a control period, shaded in red. Panel (a) shows estimates for San Francisco County, and Panel (b) shows estimates for Los Angeles County.

Figure 11: Estimated Tax Revenue Losses




Note: This figure presents a boxplot of estimated tax revenue losses in billions of dollars for Los Angeles County (in red) and San Francisco County (in blue). Tax revenue losses are estimated over a range of possible holding periods for inherited properties.

# B Tables

Table 1: Transfer Scenario Tradeoffs

| **Category** | **Transfer Before Prop 19** | **Transfer at Death** |
|---|---|---|
| Property Taxes | Lower<br>(Not reassessed) | Higher<br>(Reassessed) |
| Capital Gains Taxes | Higher<br>(No step-up in basis) | Lower<br>(Step-up in basis) |
| Parental Asset Control | Lower | Higher |

Note: This table summarizes transfer trade-offs for two scenarios: transferring property at death versus before Proposition 19 takes effect. The trade-offs are considered over three dimensions: property tax liability, capital gains liability, and parental asset control.

Table 2: Sample Averages, 2020

| | Full Sample | | Ever Transferred | |
|---|---|---|---|---|
| | SF | LA | SF | LA |
| Market Value | $1,828,906 | | $1,820,208 | $854,346 |
| Taxable Value | $929,227 | $567,626 | $735,005 | $308,129 |
| Untaxed Value | $900,564 | | $1,086,882 | $543,048 |
| Effective Tax Rate | 0.82% | | 0.63% | 0.44% |
| Years Since Sale | 20.33 | 20.55 | 22.32 | 24.61 |
| Sqft | 2234.91 | 2063.29 | 2583.03 | 1718.9 |
| Median HH Income | $145,003 | $87,207 | $139,809 | $82,057 |
| N | 193,116 | 2,152,235 | 17,063 | 62,252 |

Note: This table provides summary statistics for property values and home characteristics for properties in San Francisco and Los Angeles Counties. These values are obtained using data from the year 2020. Columns 2 and 3 provide values for the full sample of residential properties in each county, while columns 4 and 5 provide values for the subset of homes that have ever been transferred in each county.

Table 3: Summary Statistics, Window vs. Prior-Year Transfers

| | **Baseline Mean SF** | **Difference in window rel. prior year SF** | **Baseline Mean LA** | **Difference in window rel. prior year LA** |
|---|---|---|---|---|
| Effective tax rate | 0.35 | 0.05 | 0.53 | -0.15*** |
| | | (0.04) | | (0.01) |
| Assessed value (in 2020 USD) | 520,396 | 17,817 | 452,237 | 58,638* |
| | | (40,811) | | (35,131) |
| Market value (in 2020 USD) | 1,734,727 | 172,013*** | 828,563 | 574,043*** |
| | | (63,381) | | (39,374) |
| Years since last sale | 15 | 5.33*** | 20 | 8.40*** |
| | | (0.83) | | (0.75) |
| No. beds | 1 | 0.09 | 3 | 0.59*** |
| | | (0.11) | | (0.07) |
| No. baths | 3 | -0.02 | 2 | 0.65*** |
| | | (0.32) | | (0.06) |
| Property size (sq. ft.) | 2,555 | 18.97 | 1,691 | 570.64*** |
| | | (230.84) | | (46.40) |
| Median HH Income (Tract) | 139,549 | -1,934.54 | 84,478 | 8,468.51*** |
| | | (2,178.90) | | (1,087.82) |
| Observations | | 3,489 | | 4,884 |

Note: This table reports the differences in property characteristics (in Column 1) between homes transferred in the four-month transition window and homes transferred the year prior. Column 2 and 4 report the baseline averages. Column 3 and 5 report the average difference between the window transfers and non-window transfers for San Francisco and Los Angeles, respectively. Standard errors are clustered by property and reported in parenthesis. $*p < 0.1, **p < 0.05, ***p < 0.01$.

# C Appendix

## C.1 Estimating Market Value

Because Zillow coverage is incomplete for transferred properties in San Francisco, we impute missing market values. We begin by estimating property fixed effects and tract-specific linear time trends using properties with observed market values:

$$\mathrm{MV}_{it} = \alpha_i + \beta_{g(i)}\mathrm{year}_t + \varepsilon_{it}, \tag{5}$$

where MV$it$ denotes the market value of property $i$ in year $t$, $\alpha_i$ is a property fixed effect, and $\beta_{g(i)}$ is a tract-specific linear time trend for tract $g(i)$. For properties observed in some years but not others, we use the estimated fixed effect and tract trend to predict missing values.

For properties with no observed market values in our sample period, we impute $\hat{\alpha}i$ and $\hat{\beta}g(i)$ using nearest-neighbor matching. For each such property, we identify the five closest observed properties based on geographic proximity. We then compute a similarity score using standardized differences in housing and neighborhood characteristics, including distance, tract median income, bathrooms, bedrooms, rooms, units, square footage, lot area, and number of stories:

$$\mathrm{similarity}ij = \frac{1}{1+\sum_k |zik - z_{jk}|},$$

where $z_{ik}$ is the standardized value of characteristic $k$ for property $i$. We assign weights proportional to these similarity scores, $\omega_{ij} = \frac{\mathrm{similarity}_{ij}}{\sum j' \mathrm{similarity}_{ij'}}$, and use them to impute the missing components:

$$\hat{\alpha}i = \sum_j \omega ij\alpha_j, \qquad \hat{\beta}g(i) = \sum_j \omega_{ij}\beta_{g(j)}.$$

If any matched neighbor is missing one or more characteristics, we instead assign equal weights across the five neighbors. We then predict market values using equation (5) and winsorize the resulting distribution at the 5th and 95th percentiles.

## C.2 Framework Details

As summarized in Table 1, we consider a parent who decides whether or not to accelerate a transfer of property to their child based on three considerations: asset control, property taxes, and capital gains taxes. The parent has the choice to transfer property in period $t = 0$ (before Proposition 19 takes effect) or at their death in period $t = d$. We assume that the child will sell the property in period $t = s > d$ regardless of the transfer status of the property. For simplicity, we also assume a perfectly dynastic setting in which the parent values their own and their child's cash flow/utility equally.

**Asset Control**

Let the parent's utility of home consumption be denoted by $U_p$ and the child's by $U_c$, with a discount rate $\rho$. If the parent chooses to transfer the property at death, they will fully consume the house until they die. If they choose to transfer the house before death, they will still consume the house at rate $q < 1$, as dictated by parent-child negotiations. If the property is transferred at death, the child will only consume the house from the time of the parent's death to the time of sale. If the property is transferred before the parent's death, the child consumes the property at rate $1 - q$. These terms are summarized in the table below.

| | **Transfer Before Prop 19** | **Transfer At Death** |
|---|---|---|
| Parent | $q \times \sum_{t=0}^{t=d} \rho^t U_p$ | $\sum_{t=0}^{t=d} \rho^t U_p$ |
| Child | $(1-q)\sum_{t=0}^{t=d} \rho^t U_c + \sum_{t=d}^{t=s} \rho^t U_c$ | $\sum_{t=d}^{t=s} \rho^t U_c$ |

**Property Taxes**

Let the property's taxable value at time $t = 0$ be denoted by $TV_0$. Let the California state property tax rate (approximately 1%) be defined by $\tau_p$. Finally, let a property's market value at the time of the parent's death be denoted $MV_d$. In both transfer scenarios, the property's low tax basis is retained before death for $t \in [0, d)$, and increases in taxable value are limited to 2% per year. After death, property taxes continue to be capped in the transfer-before-Prop-19 scenario. However, property taxes rise sharply in the transfer-at-death scenario. This is because the property is reassessed at time $t = d$ to market value, and the child becomes liable for property taxes on this higher value. These terms are summarized in the table below.

| | **Transfer Before Prop 19** | **Transfer At Death** |
|---|---|---|
| Before Death | $\sum_{t=0}^{t=d} \rho^t \tau_p TV_0 \times 1.02^t$ | $\sum_{t=0}^{t=d} \rho^t \tau_p TV_0 \times 1.02^t$ |
| After Death | $\sum_{t=d}^{t=s} \rho^t \tau_p TV_0 \times 1.02^t$ | $\sum_{t=d}^{t=s} \rho^t \tau_p MV_d \times 1.02^{t-d}$ |

### Capital Gains Taxes

Let the capital gains tax rate be defined by $\tau_{cg}$.[14] Let $MV_s$ denote the property's market value at time of sale, and $MV_p$ denote the property's market value at time of purchase by the parent, meaning that $MV_p << MV_d$. If the property is transferred before Proposition 19 takes effect, then the child forgoes the step-up in the property's basis at the time of the parent's death. This means that their capital gains tax burden will be large relative to the transfer-at-death scenario. These terms are summarized in the table below.

| **Transfer Before Prop 19** | **Transfer At Death** |
|---|---|
| $\rho^t \tau_{cg}[MV_s - MV_p]$ | $\rho^t \tau_{cg}[MV_s - MV_d]$ |

However, note that capital gains taxes are only triggered if the asset is sold—the child may hold the property in perpetuity and not face any capital gains taxes.

### Transfer Decision

Combining terms, accelerating the property transfer will be beneficial if:

$$\sum_{t=d}^{t=s} \rho^t \tau_p [MV_d \times 1.02^{t-d} - TV_0 \times 1.02^t] > \sum_{t=0}^{t=d} \rho^t (1-q)[U_p - U_c] + \rho^t \tau_{cg}[MV_d - MV_p] \quad (6)$$

The left-hand side denotes the property tax savings from transferring the property before the policy takes effect. The first term on the right-hand side represents the additional utility the parent would receive by retaining control of the property until death. The second term on the right-hand side denotes the increase in capital gains liability if the property is transferred before death, and the child chooses to sell the property.[15] In other words, the parent's decision rule should be as follows: the parent should transfer the property before the policy if the property tax savings from avoiding reassessment outweigh their concerns about asset control plus the potential future increase in capital gains taxation.

Assuming the utility of home consumption is the same for parent and child, we visually demonstrate the trade-off between the cash flow of property taxes and capital gains over

[14]Capital gains are taxed at a rate of 0%, 15%, or 20% depending on the filer's income. There is a $250,000 per filer capital gains exclusion for the sale of your primary home, which we choose to omit here.

[15]If the child never plans to sell the property, this term is equal to zero.

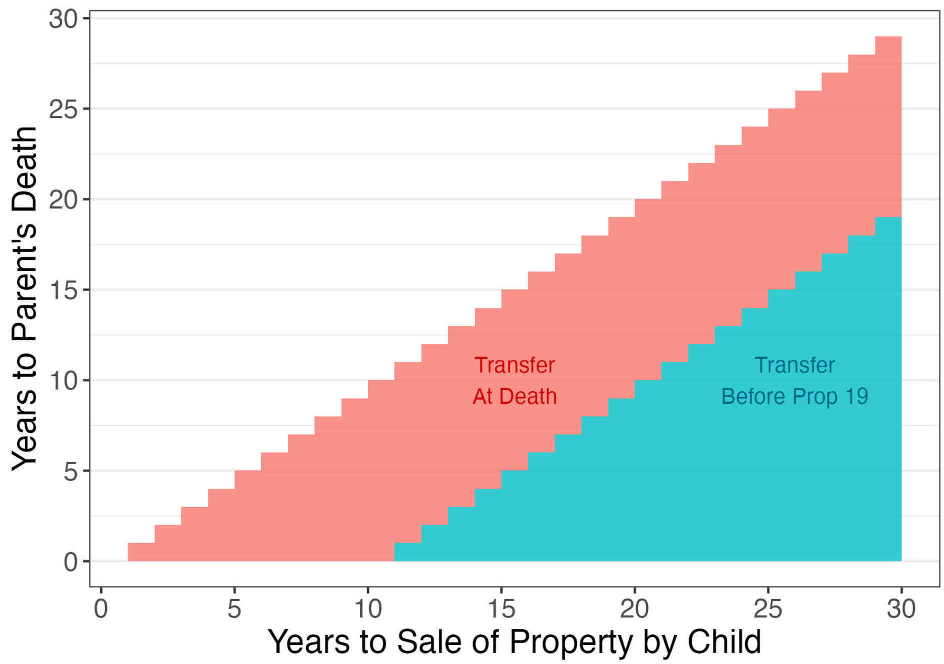
different time horizons of parental death and child's length of holding in Figure A1.[16]

| **Term** | **Description** | **Value** |
|---|---|---|
| $\rho$ | Discount rate | 0.95 |
| $\tau_p$ | Property tax rate | 0.01 |
| $\tau_{cg}$ | Capital gains tax rate | 0.15 |
| $MV_p$ | Property's purchase price | \$500,000 |
| $a$ | Property appreciation rate, such that $MV_t = MV_p(1+a)^t$ | 0.07 |
| $d$ | Years until parent's death | $[0,30]$ |
| $s$ | Years until property's sale, $s > d$ | $[0,30]$ |

Using the parameter values above, the simulation reveals that an accelerated transfer is most beneficial if the child plans to hold the property for a significant length of time ($s-d > 10$), and/or if the parent is close to death. Moreover, we find that the longer the child plans on holding the property, the larger the potential tax savings relative to the transfer-at-death scenario; the periods of low property taxes outweigh the future increase in capital gains liability. Similarly, if the parent will die in the near future, accelerating the transfer shields the child from many periods of high property taxation, which accumulates enough savings to outweigh higher capital gains liability.

[16]Figure A5 repeats this exercise in the primary-to-primary home transfer scenario, which allows for the exclusion of \$1m in market value. This changes Equation (6) to $\sum_{t=d}^{t=s} \rho^t \tau_p[(MV_d - 1m) \times 1.02^{t-d} - TV_0 \times 1.02^t] > \sum_{t=0}^{t=d} \rho^t (1-q)[U_p - U_c] + \rho^t \tau_{cg}[MV_d - MV_p]$.

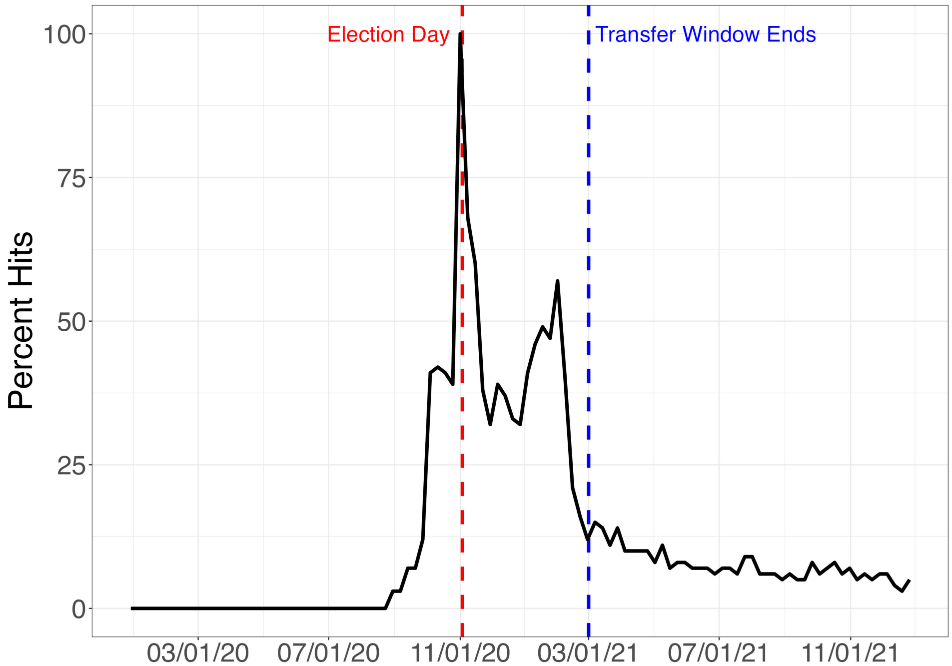
Figure A1: Gains to Property Transfer, Capital Gains Liability vs. Property Tax Savings




Note: This figure shows the conditions under which it is beneficial for a parent to transfer property at death (shaded in red) versus transfer property before Proposition 19 takes effect (shaded in blue). Time-of-transfer optimality is dictated by the number of years the child expects to hold the property (x-axis) and the number of years until the parent's death (y-axis).

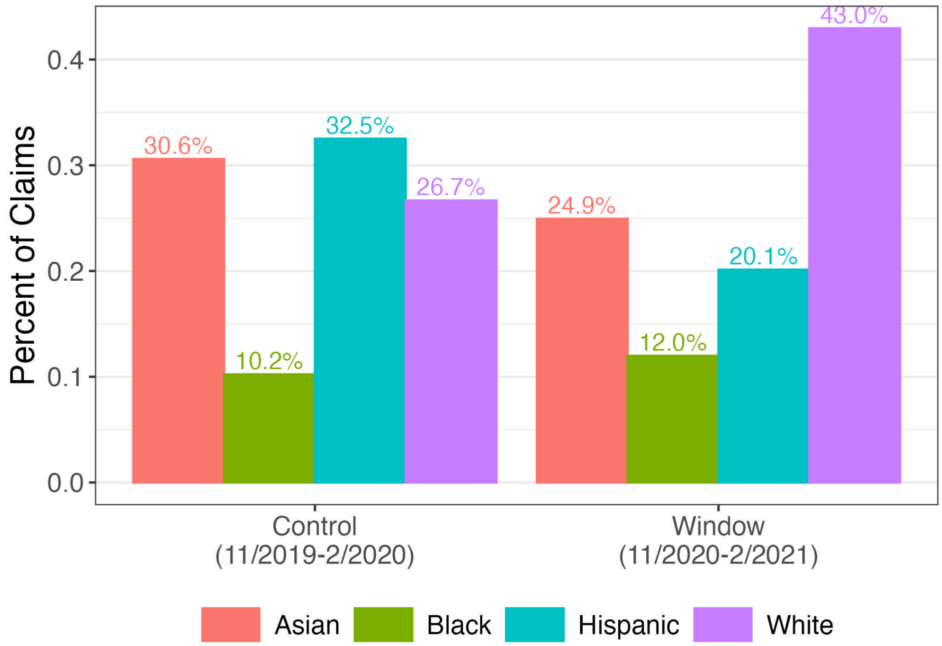
## C.3 Figures

Figure A2: Google Trend for "California Proposition 19, Property Tax Transfers, Exemptions, and Revenue for Wildfire Agencies and Counties Amendment (2020)"



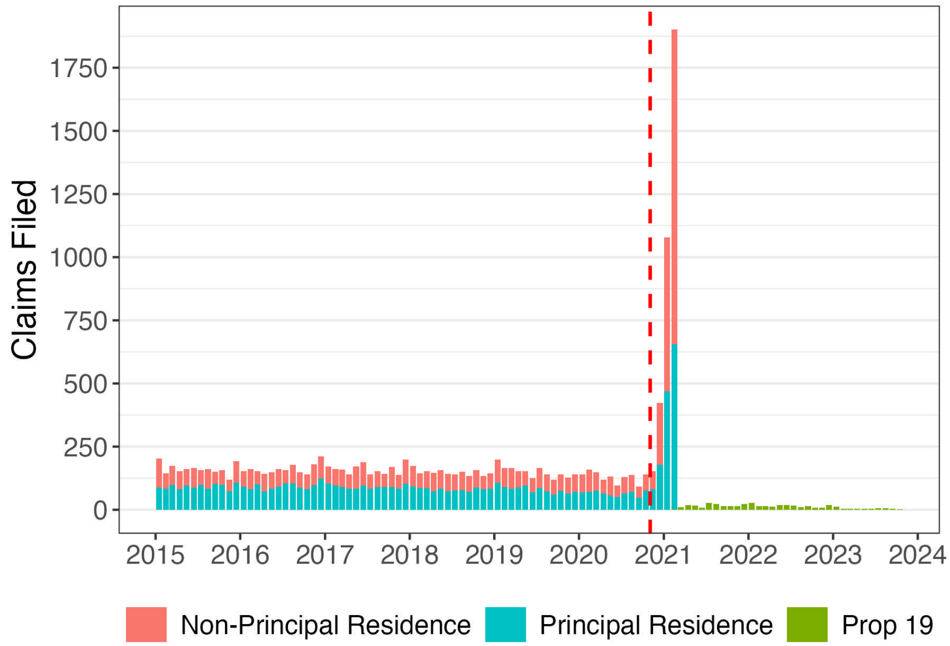

Note: This figure plots weekly Google searches of "California Proposition 19, Property Tax Transfers, Exemptions, and Revenue for Wildfire Agencies and Counties Amendment (2020)" from January 2020 to December 2021. The red dashed vertical line marks Election Day 2020, and the blue dashed vertical line marks the date Proposition 19 took effect.

Figure A3: Percent of Claims by Predicted Owner Race in Los Angeles County



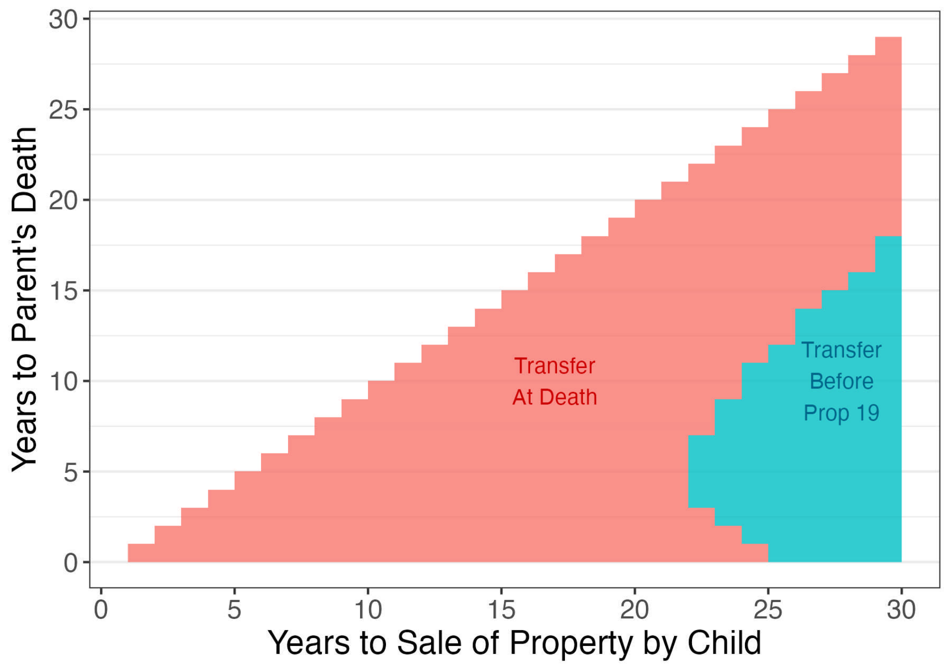

Note: This figure plots the percentage of property transfer claims by racial group, shown separately for the treatment ('Window') period and a control period. Racial group is estimated by owner name using an algorithm from the `rethnicity` package.

Figure A4: Parent-child Transfers in San Francisco County, Pre- and Post-Proposition 19




Note: This figure plots the number of unique parent-to-child property transfers by month from 2015 to 2024 in San Francisco County. The red bars represent the number of claims for non-principal or secondary homes and the teal bars represent the number of claims for principal or primary homes. The red dashed line denotes the passage of Proposition 19. The green bars represent the number of transfers occurring after Proposition 19, which are primary residence to primary residence transfers only.

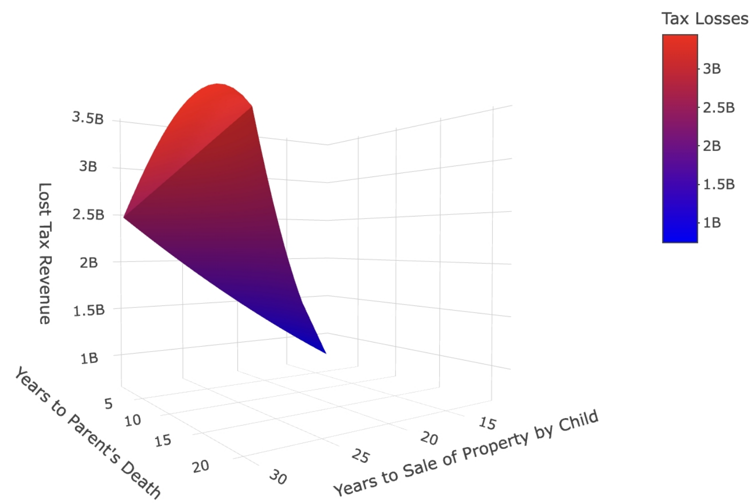
Figure A5: Gains to Property Transfer, Capital Gains Liability vs. Property Tax Savings, Primary-to-Primary Home Transfer



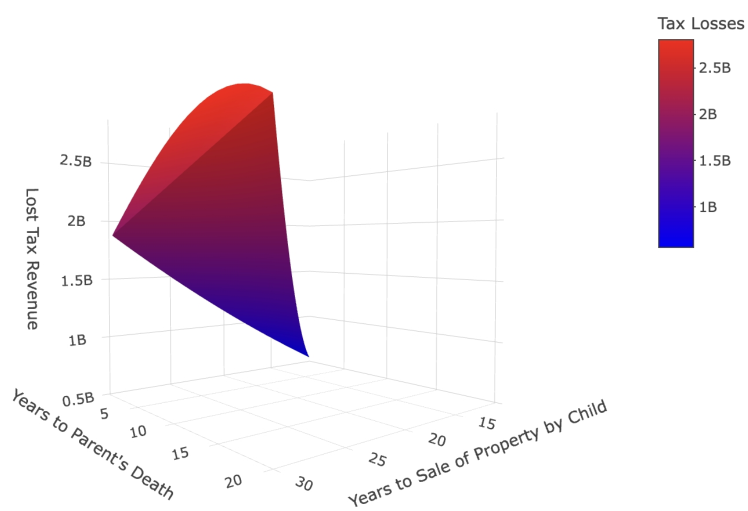

Note: This figure shows the conditions under which it is beneficial for a parent to transfer property at death (shaded in red) versus transfer property before Proposition 19 takes effect (shaded in blue) for primary-to-primary home transfers, which continue to qualify for a $1m exemption after the passage of Proposition 19. Time-of-transfer optimality is dictated by the number of years the child expects to hold the property (x-axis) and the number of years until the parent's death (y-axis).

Figure A6: Full Range of Estimated Tax Revenue Losses

(a) San Francisco County




(b) Los Angeles County




Note: This figure plots the full estimated range of property tax revenue losses by 1) the number of years until the parent's death and 2) the number of years the child-inheritor holds the inherited property. Panel (a) provides estimates for Los Angeles County, and Panel (b) shows estimates for San Francisco County. In both panels, smaller losses are shaded in blue, while larger losses are shaded in red.

## C.4 Intertemporal Elasticity

### Method

We estimate the elasticity of home transfers to a change in the property tax rate in three ways.

First, we calculate the elasticity with respect to the raw change in claims:

$$e_{Claims} = \frac{Claims_W - Claims_C}{Claims_C} \times \frac{1}{\log(1-\tau_0) - \log(1-\tau_1)} \tag{7}$$

where $Claims_W$ are claims made during the transfer window of November 2020–February 2021; $Claims_C$ are claims made during a control period of November 2019–February 2020; $\tau_0$ is the effective tax rate if the property is transferred before the policy change; and $\tau_1$ is the effective tax rate if the property is transferred right after the policy change. To calculate the number of "excess" claims or bunchers, we compare the transfer window of November 2020–February 2021 with a control period of November 2019–February 2020. Given the stability of claims over time, we rely on the average number of claims across all periods rather than fitting a high-order polynomial. This approach avoids the interpretational challenges associated with seventh-degree polynomials used in some other studies, while yielding comparable results in our setting.

Second, we estimate the elasticity with respect to the market value of transferred homes:

$$e_{MV} = \frac{MV_W - MV_C}{MV_C} \times \frac{1}{\log(1-\tau_0) - \log(1-\tau_1)} \tag{8}$$

where $MV_W$ are the market values of the claims made during the transfer window of November 2020–February 2021; $MV_C$ are the market values of claims made during a control period of November 2019–February 2020. We prefer this specification for ease of comparison to other elasticity estimates, which are mostly focused on the dollar value of transferred assets instead of changes in the volume of transfers.

Third, we estimate the elasticity with respect to the "Tax Base" is as follows:

$$e_{THS} = \frac{Claims_W - Claims_C}{TaxBase} \times \frac{1}{\log(1-\tau_0) - \log(1-\tau_1)} \tag{9}$$

Here we normalize the change in claims to the estimated tax base, which we define as the estimated number of residential properties eligible for transfer. We calculate the estimated tax base by restricting to 85% of the residential housing stock owned for over 20 years.[17] This represents the percent change in eligible housing stock transferred due to the policy change.

To calculate the percent change in the net-of-tax rate due to the policy change, we define $\tau_0$ as the pre-transfer effective property tax rate:

$$\tau_0 = \frac{\tau_p \times TaxableValue}{MarketValue} = \frac{PropertyTaxPayment}{MarketValue}$$

which is equivalent to the tax rate retained if the home is transferred before the deadline. $\tau_1$ equals the tax rate if the home is transferred after the deadline and thus reassessed to market value, meaning

$$\tau_1 = \frac{\tau_p \times MarketValue}{MarketValue} = \tau_p$$

which is equivalent to approximately 1% in California, with small variations in specific counties and cities.

**Results**

We report the estimated elasticities in Tables A2-A4. The sample average effective tax rates are presented under $\tau_0$ of each table. Because each individual homeowner faces a unique real tax rate, we use the average effective tax rate for the window sample shown of each table to calculate the elasticities in Equations (2)-(4), which are presented in the last column of each table.

Table A2 displays the elasticity of transfer claims with respect to the net of tax rate (Equation (2)). The elasticity is extremely large for both San Francisco, $e = 675$, and Los Angeles, $e = 352$. Table A3 displays the elasticity of the market value of transferred homes with respect to the net of tax rate (Equation (3)). The calculated elasticity is even larger than the claim elasticity, with values of $e = 781$ and $e = 714$ for San Francisco and Los Angeles respectively. These estimates are calculated as a direct comparison to other estimates produced in the literature, which we discuss in the next section.

Table A4 displays the elasticity of transfer claims normalized to the estimated tax base with respect to the net of tax rate (Equation (4)). The calculated elasticities are much smaller, with values of $e = 4.3$ and $e = 0.53$ for San Francisco and Los Angeles respectively. Normalization may be appropriate because responses may be concentrated among relatively few households

[17] Restricting to homes owned for at least 20 years proxies for parent and child age, and the CDC and U.S. Census suggest that 80-85% of women older than 50 have children.

utilizing this policy; hence, even minor deviations from low base rates can prompt large proportional responses. However, the elasticity estimate for San Francisco remains notably large compared to the existing literature.

#### Discussion

Our elasticity estimates, while large, may represent a lower bound. While Proposition 19 substantially increased property tax liabilities for inherited properties, another relevant tax consideration is the capital gains tax[18]. As discussion above, heirs who inherit property at death receive a "step-up" in cost basis to the property's current market value, thereby eliminating accumulated capital gains at the parent's death for tax purposes. By contrast, property transferred inter vivos does not qualify for a step-up, and the recipient inherits the original purchase price as their cost basis. As a result, accelerating transfers to avoid higher property taxes increases potential capital gains tax liabilities upon future sale. This trade-off dampens the incentive to accelerate transfers, as recipients weigh the immediate savings in property taxes against the prospect of higher capital gains taxes in the future.

Table A1 places our estimates in the context of other elasticities produced by the inheritance tax literature. Elasticities classified as "permanent" are produced by studies estimating the responsiveness of the entire tax base to a notched or kinked tax schedule. These elasticities tend to be quite small, $e < 1$ (Kopczuk (2013); Goupille-Lebret and Infante (2018); Glogowsky (2021)). One might consider the result of Equation (4) as most comparable to the estimates produced by these papers, representing the portion of the entire tax base that responds to a temporarily available incentive. Our estimates are still at the high end of these permanent elasticities, especially the estimate $e = 4.3$ from San Francisco County.

Elasticities classified as "transitory" are produced by studies estimating the temporary response to an anticipated tax rate change (Locks (2024); Joulfaian and McGarry (2004); House and Shapiro (2008)). These estimates tend to be much larger than the permanent elasticities, with estimates (excluding ours) ranging from $e = [8, 148]$. It is unsurprising that our estimate is most similar to the largest estimate (excluding ours) of $e = 148$ from Locks (2024), as our settings are the most comparable—in both cases, a large, salient, anticipated tax change took effect a few months after the tax increase was announced.

[18]The estate tax and gift taxes apply as well. However, as explained in prior sections, these taxes are not relevant for the majority of homeowners.

## Figures and Tables

Table A1: Elasticity Estimate Comparison

| **Authors** | **Context** | **Horizon** | *e* |
|---|---|---|---|
| Kopczuk (2013) | Meta-study on estate size elasticity at death | Permanent | 0.1-0.2 |
| Goupille-Lebret and Infante (2018) | Notched inheritance tax schedule by age on retirement income | Permanent | 0.23-0.36 |
| Glogowsky (2021) | Kinked inheritance tax schedule | Permanent | 0.1 |
| Joulfaian and McGarry (2004) | Future increase in gift tax | Transitory | 8.4 |
| House and Shapiro (2008) | Temporary investment tax incentives (bonus depreciation) | Transitory | 6-14 |
| Locks (2024) | Future increase in inheritance taxes | Transitory | 148 |
| Baker and Kim (2025) | Future increase in property inheritance taxes | Transitory | >700 |

Note: This table displays estimates of the elasticity of estate size/inheritance/investment to the net-of-tax-rate for a selection of relevant papers.

Table A2: Elasticity of Transfer Claims to Tax Rate Change

| | $\tau_0$ | | | Claims | | e |
|---|---|---|---|---|---|---|
| County | Sample | Control | Window | Control | Window | Claims |
| SF | 0.63 | 0.43 | 0.47 | 551 | 3294 | 675 |
| LA | 0.45 | 0.63 | 0.45 | 1100 | 3892 | 352 |

Note: This table provides estimates of the elasticity of property transfer claim volume with respect to the net-of-tax rate. Columns 2-4 display effective tax rates for the sample of ever-transferred homes, the control group, and the treatment group, respectively. Columns 5-6 provide the property transfer claim volumes for the control and treatment group, respectively. The final column provides the elasticity estimate.

Table A3: Elasticity of Transferred Home Value to Tax Rate Change

| | $\tau_0$ | | | Market Value (Bn) | | e |
|---|---|---|---|---|---|---|
| County | Sample | Control | Window | Control | Window | Market Value |
| SF | 0.63 | 0.43 | 0.47 | 0.96 | 6.48 | 781 |
| LA | 0.45 | 0.63 | 0.45 | 0.91 | 5.62 | 714 |

Note: This table provides estimates of the elasticity of the market value of property transfer claims with respect to the net-of-tax rate. Columns 2-4 display effective tax rates for the sample of ever-transferred homes, the control group, and the treatment group, respectively. Columns 5-6 provide the market value of property transfer claims for the control group and the treatment group, respectively. The final column provides the elasticity estimate.

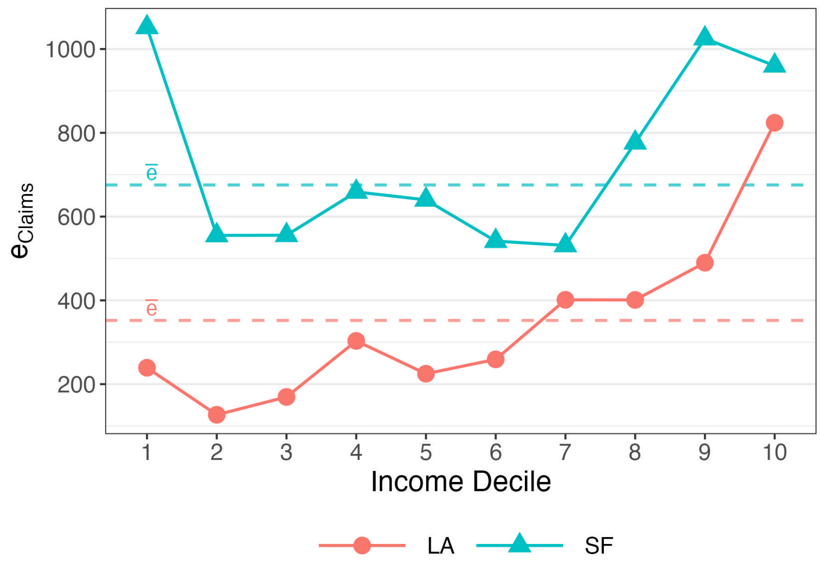
Table A4: Elasticity of Tax Base to Tax Rate Change

| | $\tau_0$ | | | Claims | | | e |
|---|---|---|---|---|---|---|---|
| County | Sample | Control | Window | Control | Window | N Base | Tax Base |
| SF | 0.63 | 0.43 | 0.47 | 551 | 3294 | 86425 | 4.3 |
| LA | 0.45 | 0.63 | 0.45 | 1100 | 3892 | 718718 | 0.53 |

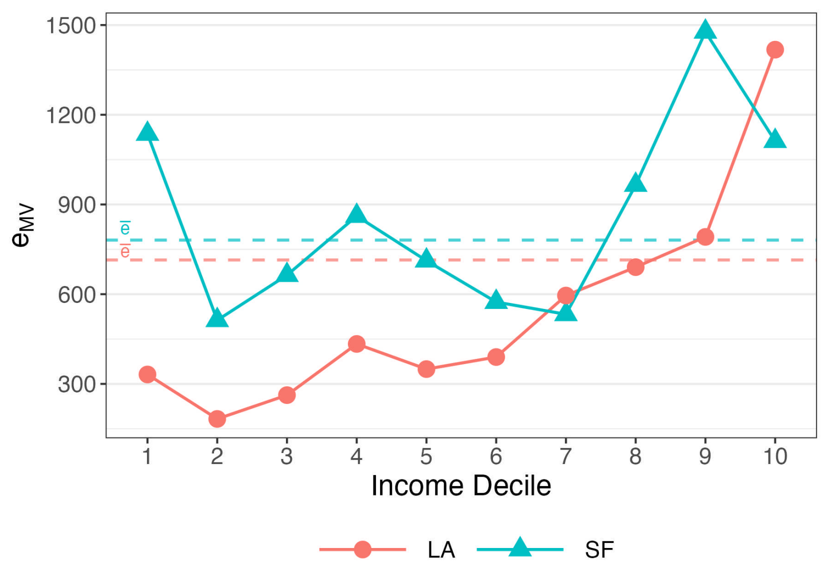
Note: This table provides estimates of the elasticity of property transfer claim volume normalized to the estimated tax base with respect to the net-of-tax rate. Columns 2-4 display effective tax rates for the sample of ever-transferred homes, the control group, and the treatment group, respectively. Columns 5-6 provide the property transfer claim volumes for the control and treatment group, respectively. Column 7 provides an estimate of the size of the tax base, calculated as 85% of homes owned for 20+ years. The final column provides the elasticity estimate.

Figure A7: Elasticity to tax rate change by income decile

(a) Claims Elasticity




(b) Market Value Elasticity




Note: This figure plots elasticity estimates by within-county income decile. Panel (a) shows the elasticity of the number of transfer claims with respect to the net-of-tax rate by income decile, while Panel (b) plots the elasticity of the market value of transferred properties with respect to the net-of-tax rate by income decile. Income decile is calculated using median Census tract income for the sample of ever-transferred homes in San Francisco and Los Angeles County.